Metro Site Access

UI Modernisation Scope Document

Commercial in confidence

Document Purpose

The purpose of this document is to outline the scope of the UI modernisation project, which involves migrating the Metro Site Access Application from AngularJS 1.x to ReactJS.

Document Information

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Division	Business Systems Software - Corporate Applications - OCMS
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Version History

Version	Date	Author	Description
V0.1	19/02/2026	Wu Ren	Initial draft.

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Name	Position	Date
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Soeren Lange	OCMS Project Manager	11/03/26

Document Approval

Name	Position	Date (via email)

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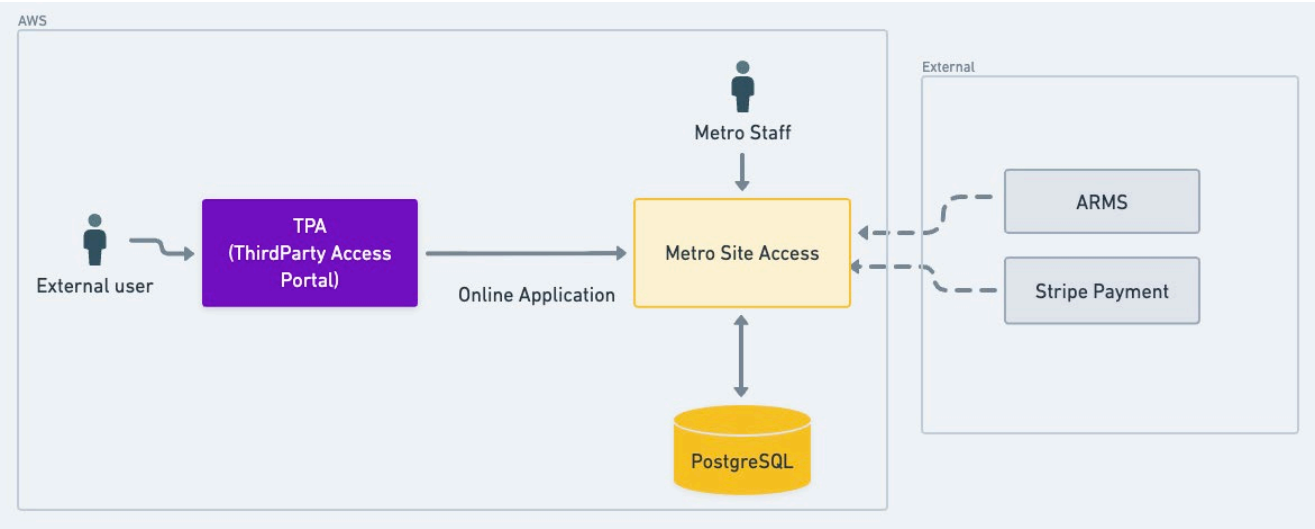
1. Introduction

1.1. Background

Business Context:

Metro Site Access provides the administrative features to administer incoming Network Corridor Access Applications. This app ensures access to the network is centrally managed and risks to operations, MTM staff, external personnel and passengers are mitigated.



External users (non-Metro personnel) typically submit their applications to access any location of the track network via the Third-Party Access Portal, after which the request progresses to Metro Site Access for standard assessment and approval workflow.

Scope Summary:

The Metro Site Access Application's current user interface is built on AngularJS 1.x, a framework that has reached end-of-life and is no longer actively supported. To address the associated technical risks and ensure long-term maintainability, the UI will undergo a modernisation initiative.

This project involves migrating the existing AngularJS 1.x interface to ReactJS. The scope is strictly limited to a like-for-like replacement of the frontend framework.

There will be:

- No new functionality introduced
- No changes to backend services
- No infrastructure or architectural modifications

The objective is solely to enhance the application's sustainability by transitioning to a modern, supported frontend technology while preserving all existing behaviour.

1.2. Glossary

The terms defined in this document are as follows:

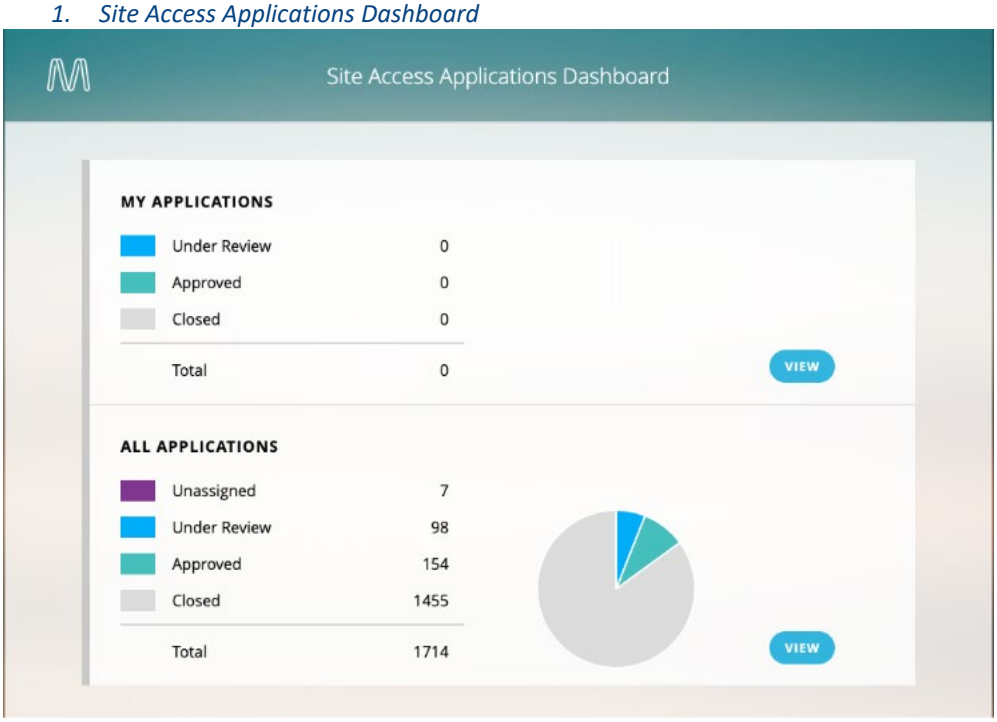
Acronym / Term	Definition
MTM	Metro Trains Melbourne
BSS	Business Support Systems
Metro Apps	InForm, OnTrack, Performance Reporting System, Metro Connect, Metro Access, Third Party access and Stations Dashboard.

2. Scope of Change

2.1. Scope Inclusions

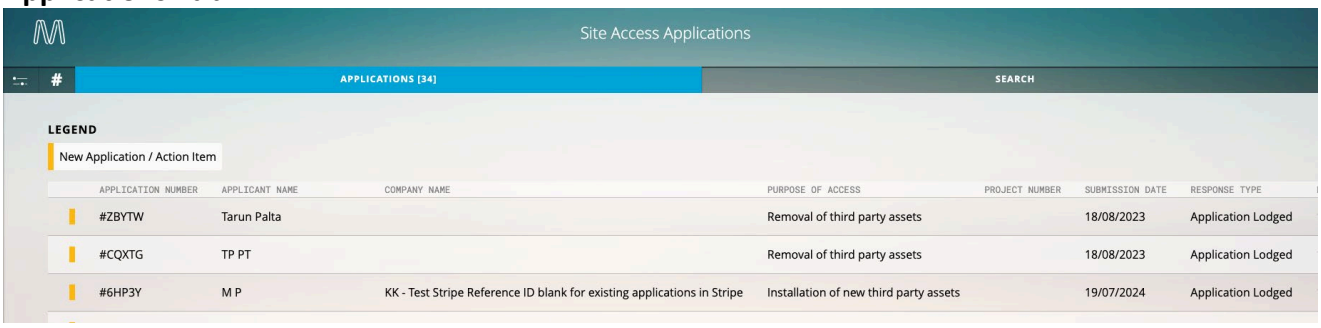
2.1.1. AngularJS to ReactJS Rewrite

1. *Site Access Applications Dashboard*



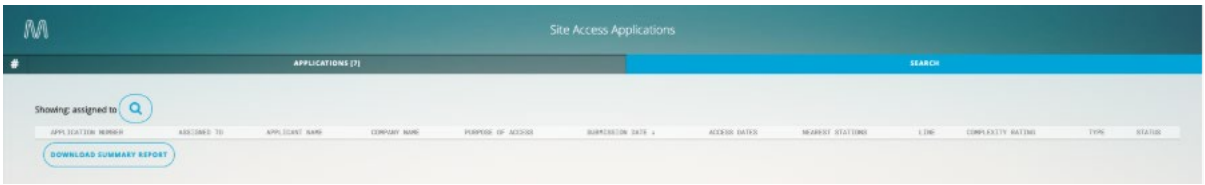
2. *Site Access Applications*

Applications Tab

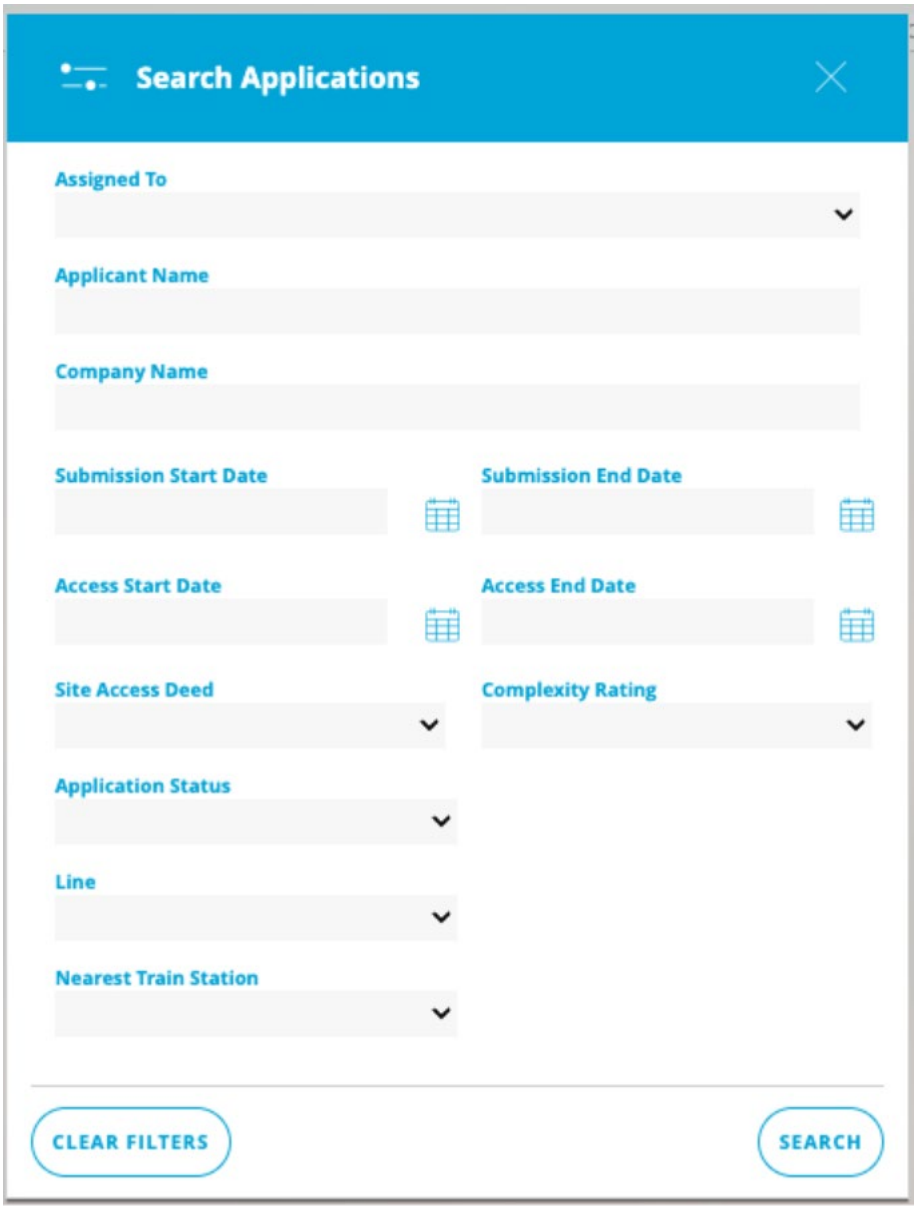


Find Application Modal

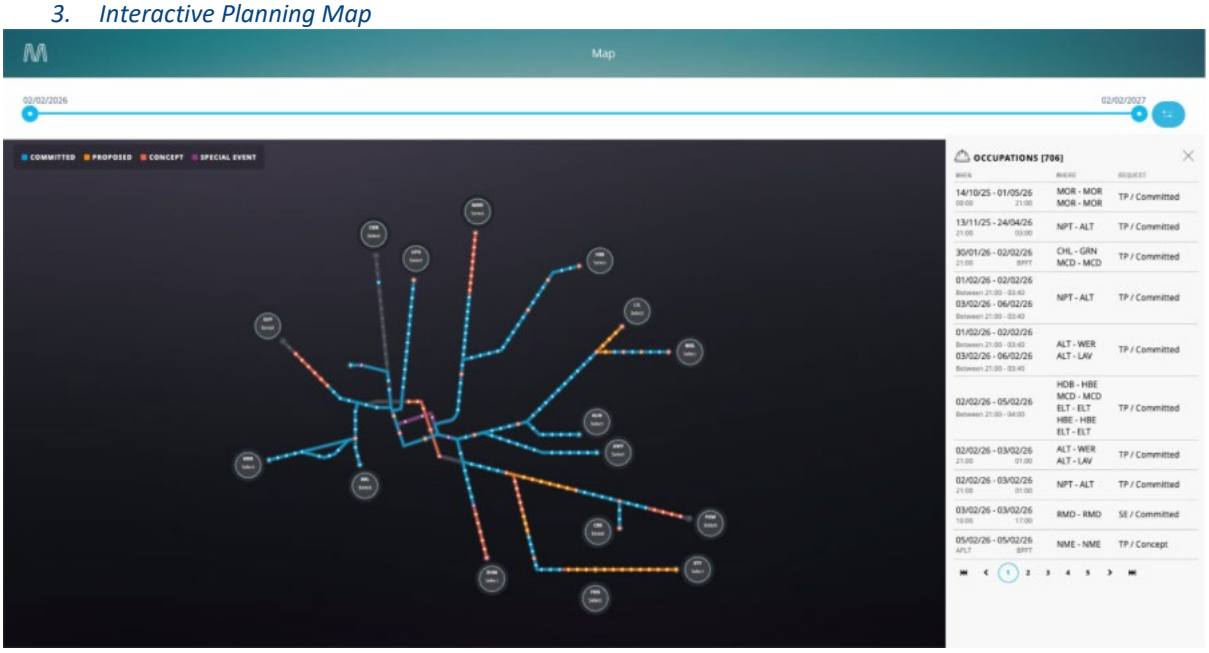
Search Tab



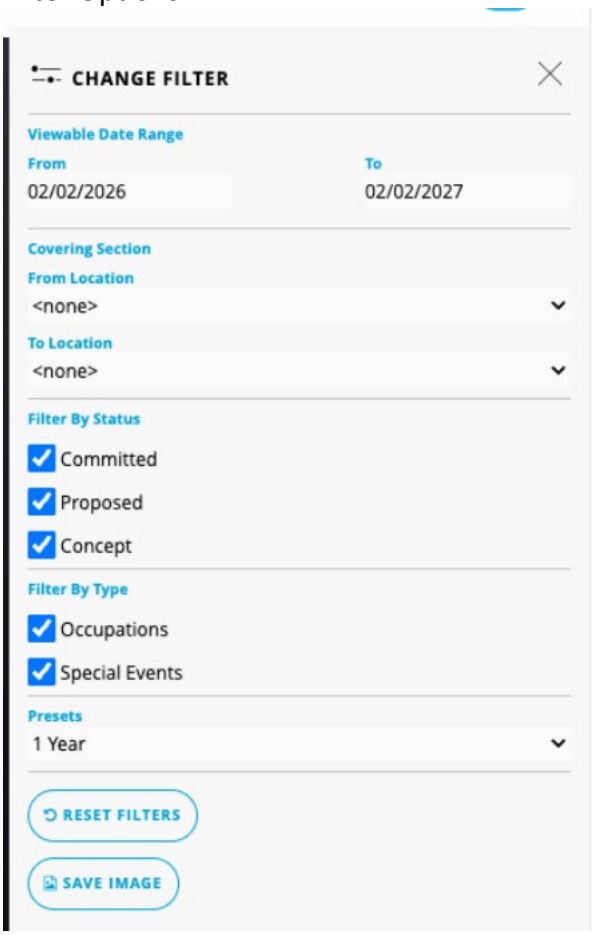
Search Modal



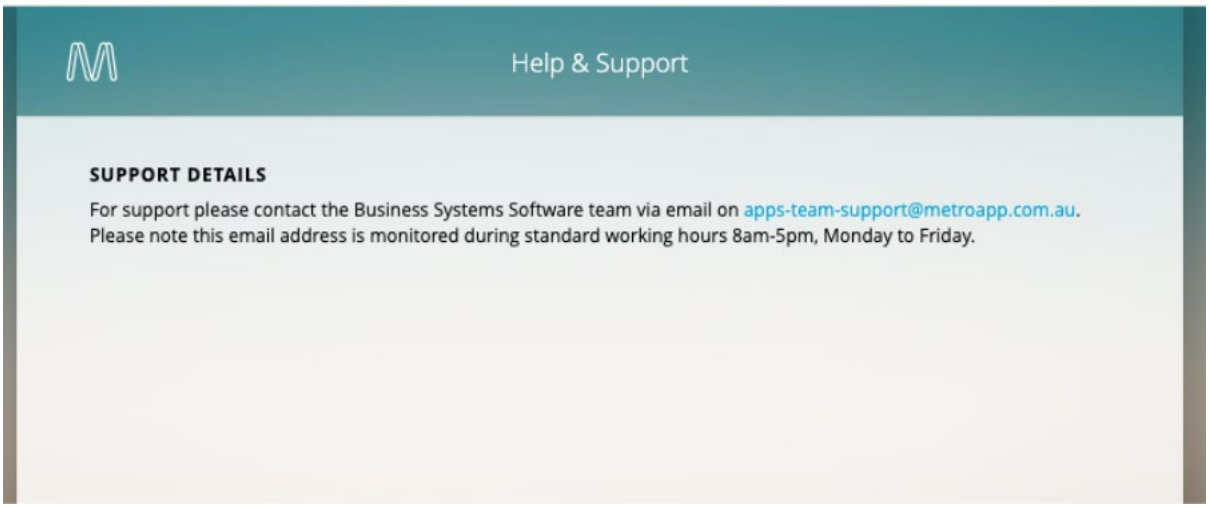
3. *Interactive Planning Map*



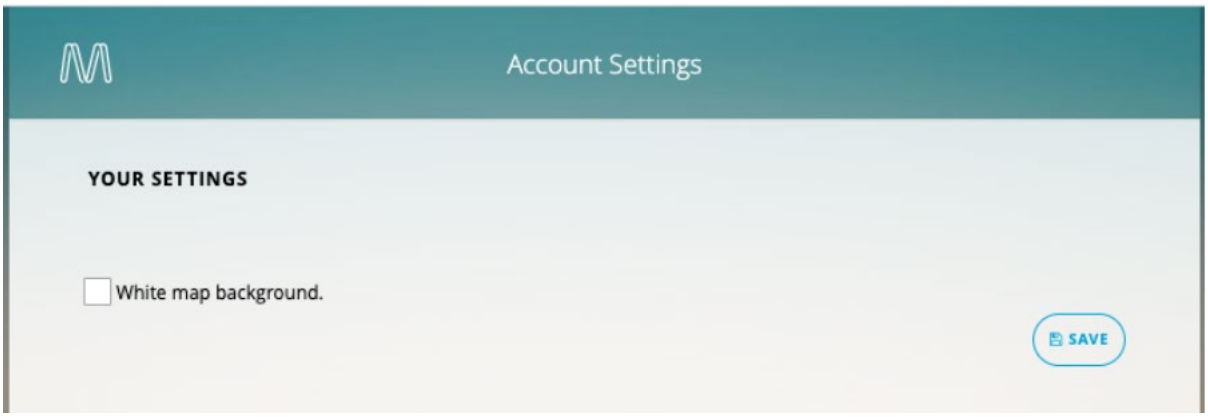
Filter Options



4. *Help & Support*



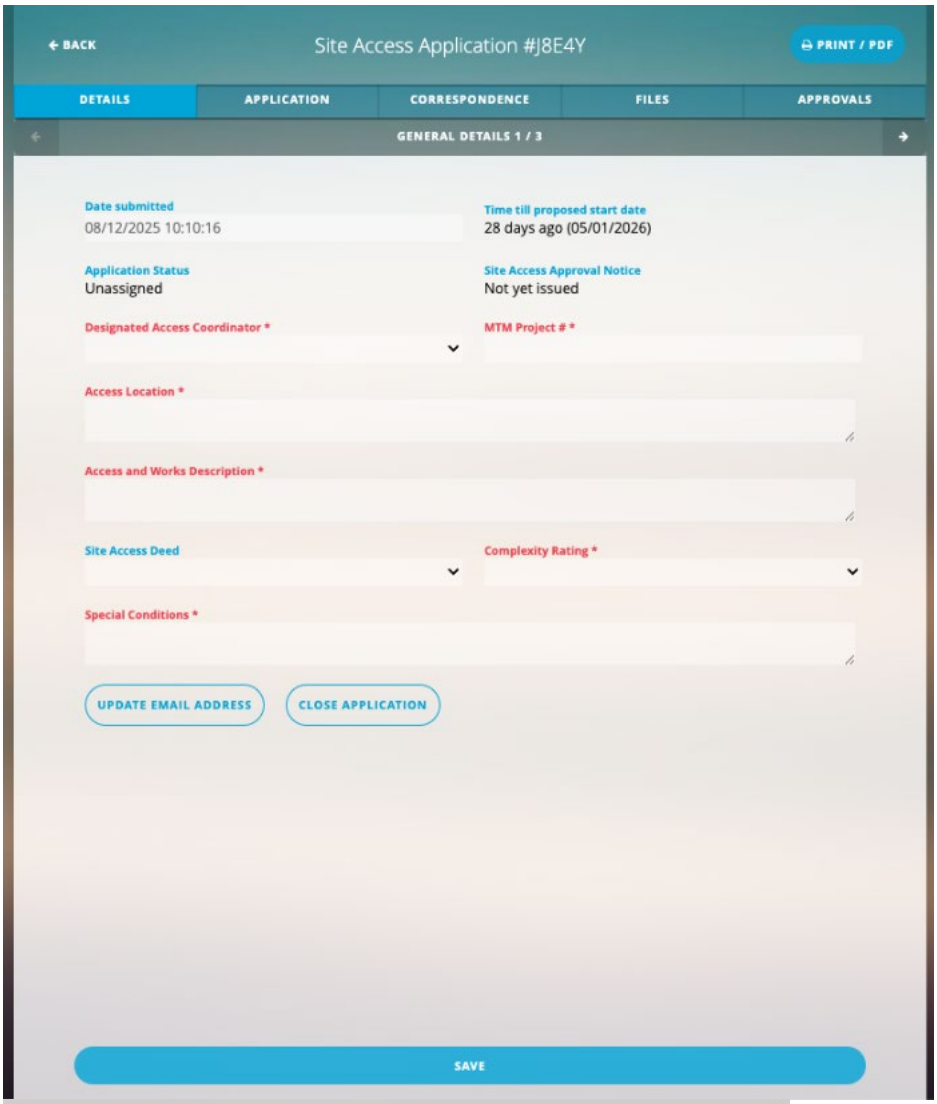
5. *Account Settings*

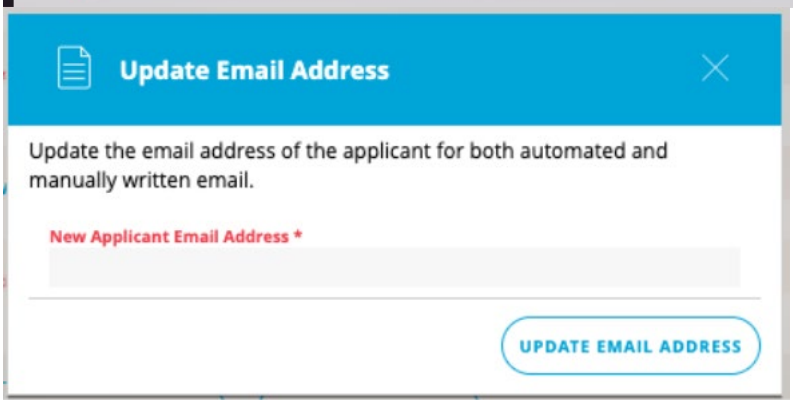


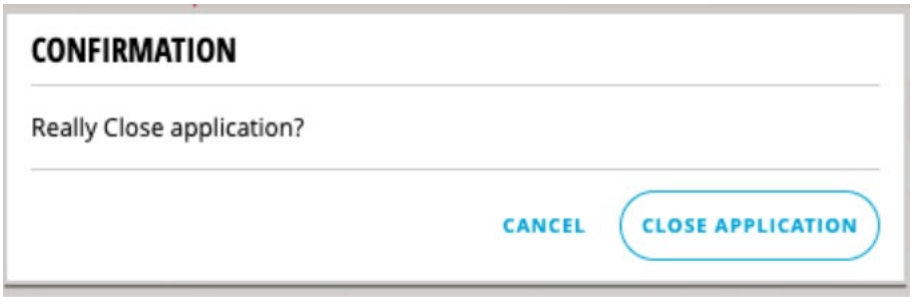
6. Site Access Application

6.1 Details Tab

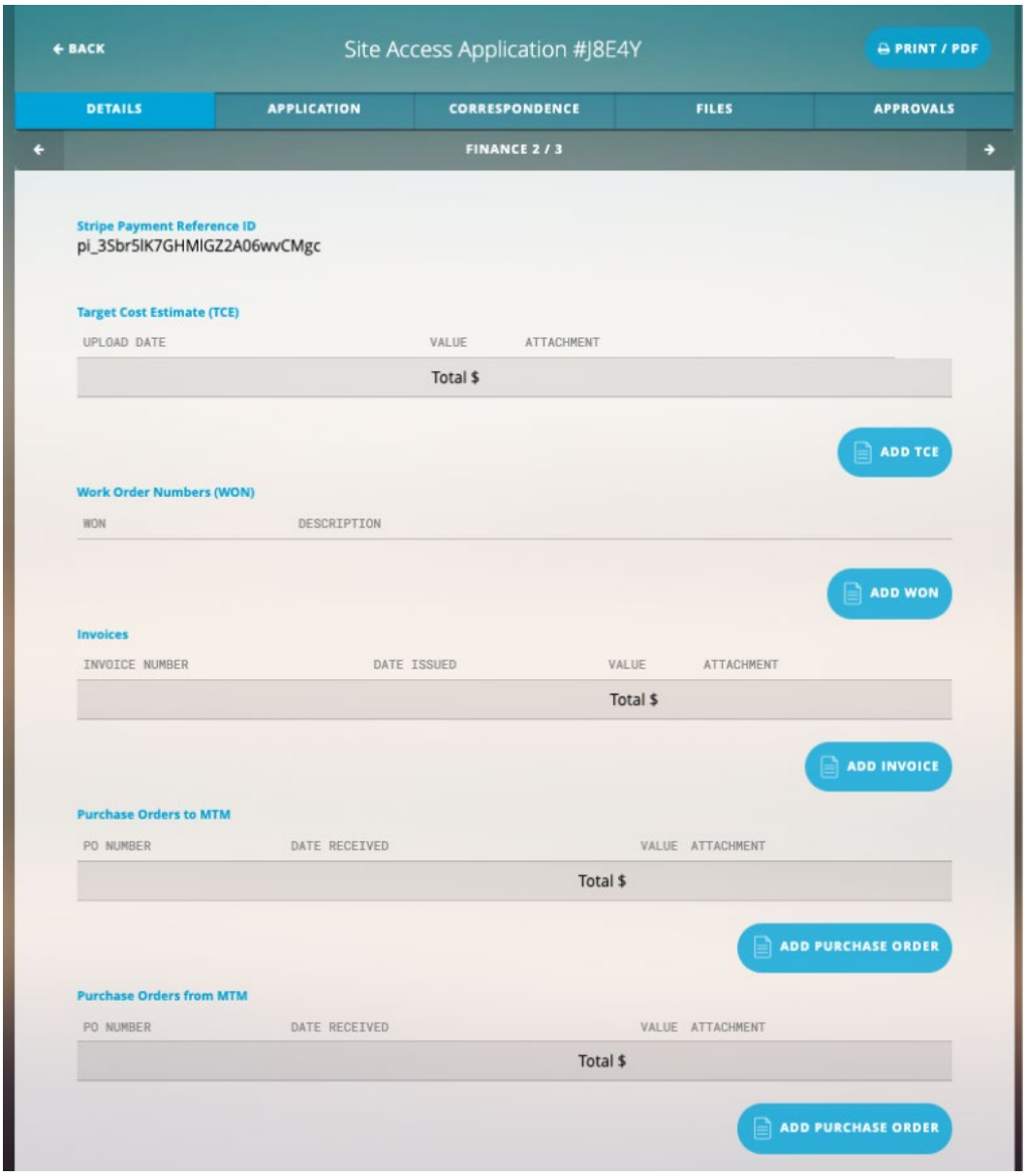
6.1.1 General Details Section

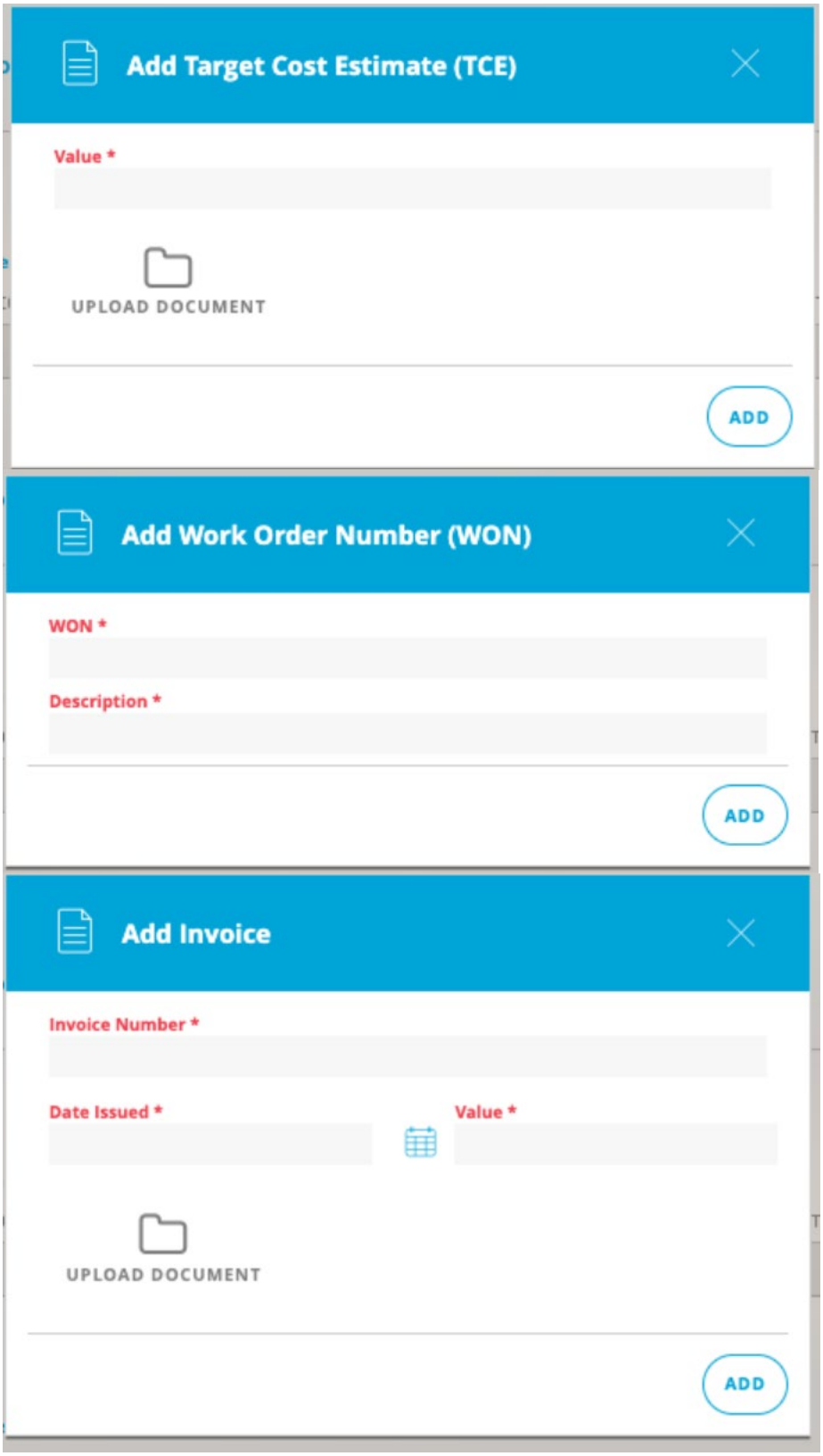


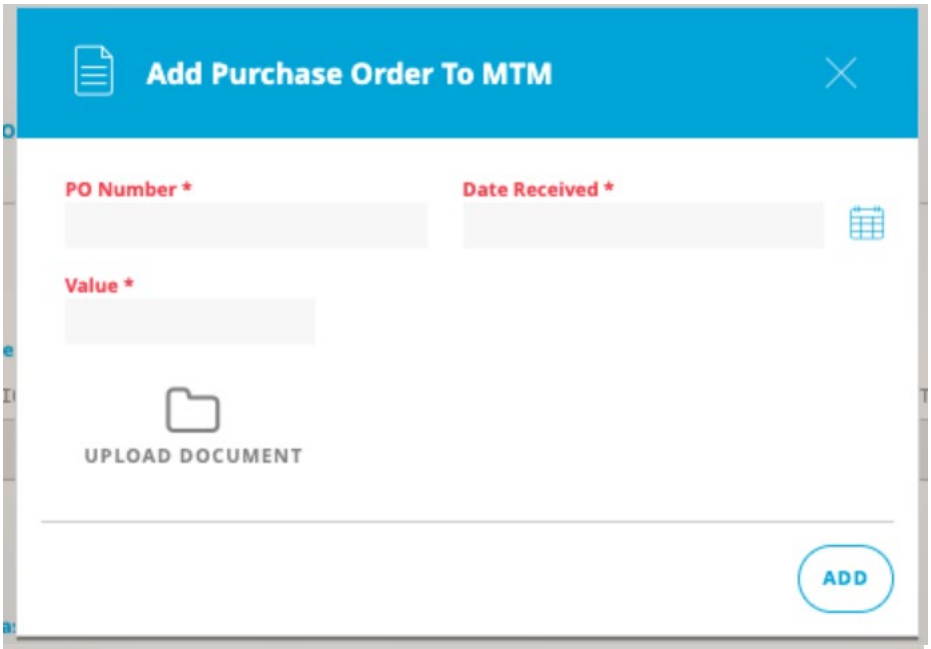


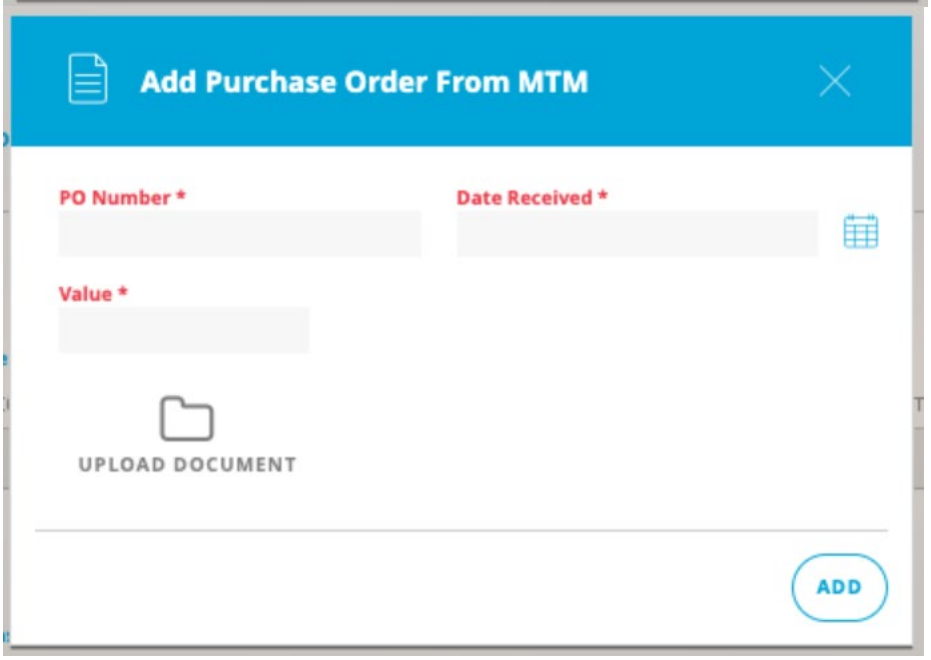


6.1.2 Finance Section

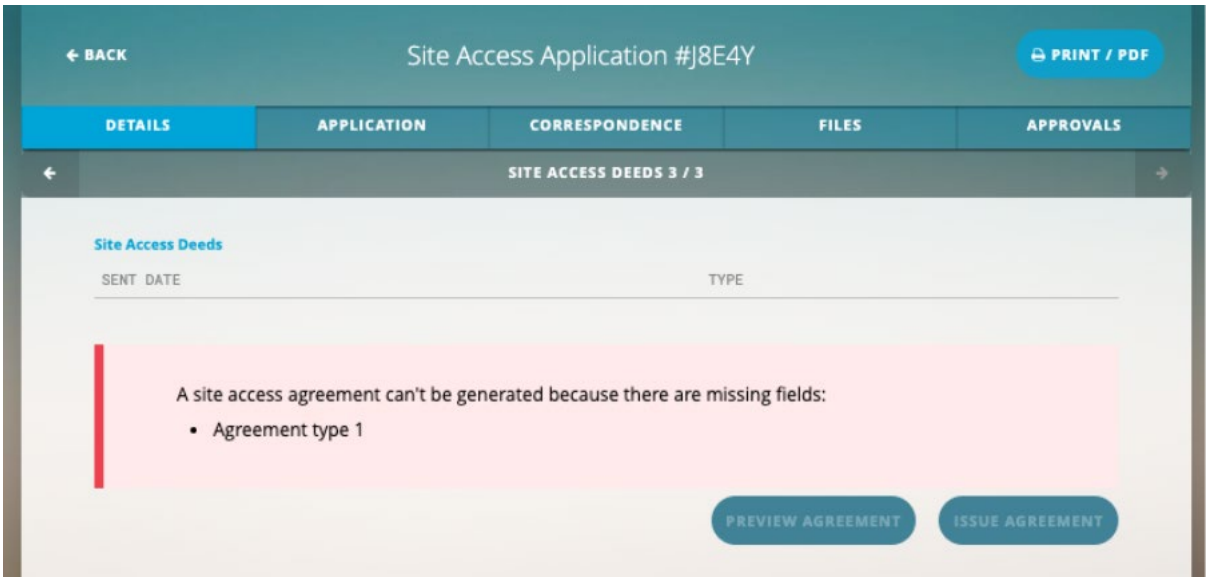



Add Target Cost Estimate (TCE)
Value *
UPLOAD DOCUMENT
ADD
Add Work Order Number (WON)
WON *
Description *
ADD
Add Invoice
Invoice Number *
Date Issued *
Value *
UPLOAD DOCUMENT
ADD


Add Purchase Order To MTM
PO Number *
Date Received *
Value *
UPLOAD DOCUMENT
ADD

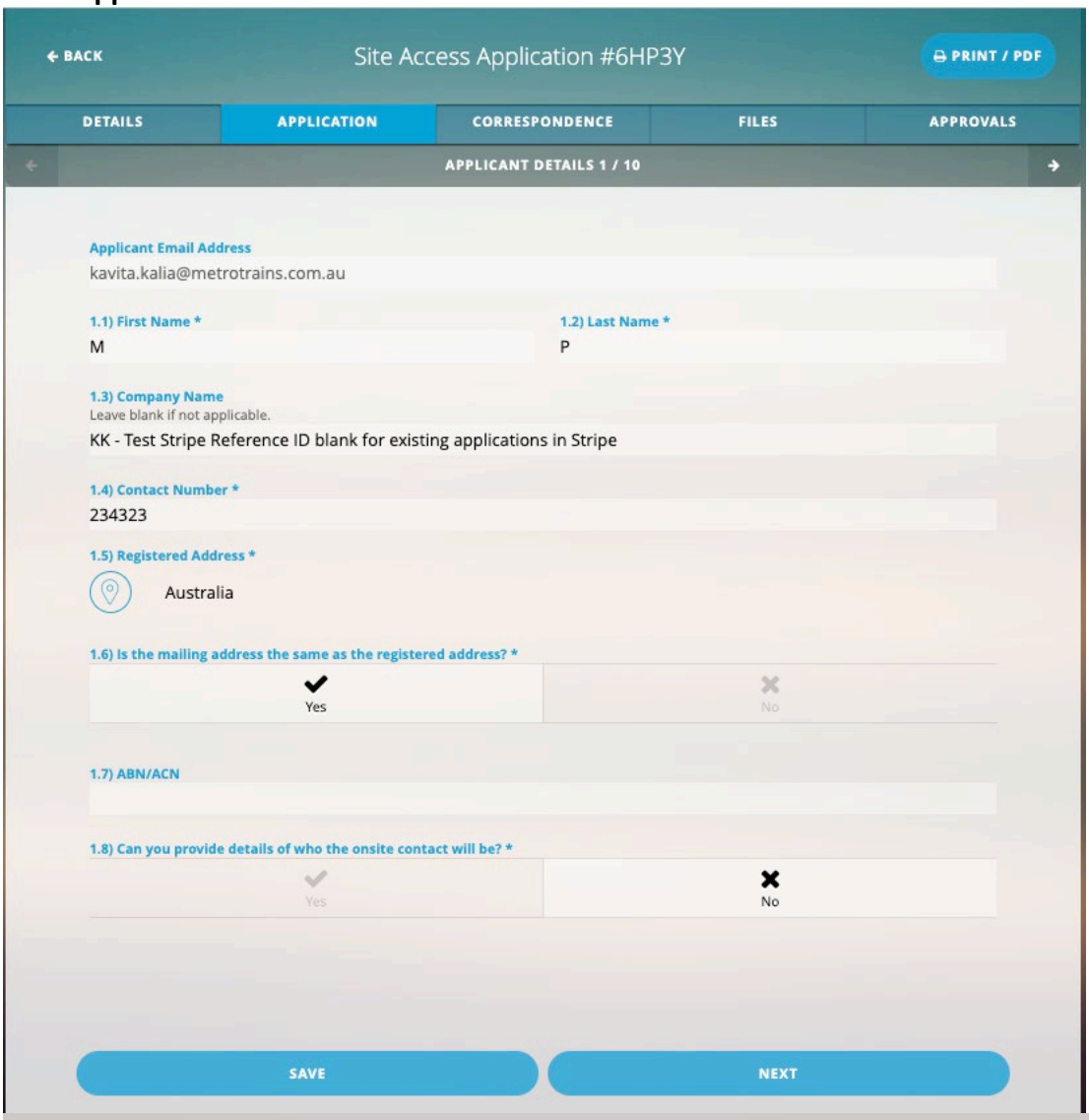

Add Purchase Order From MTM
PO Number *
Date Received *
Value *
UPLOAD DOCUMENT
ADD

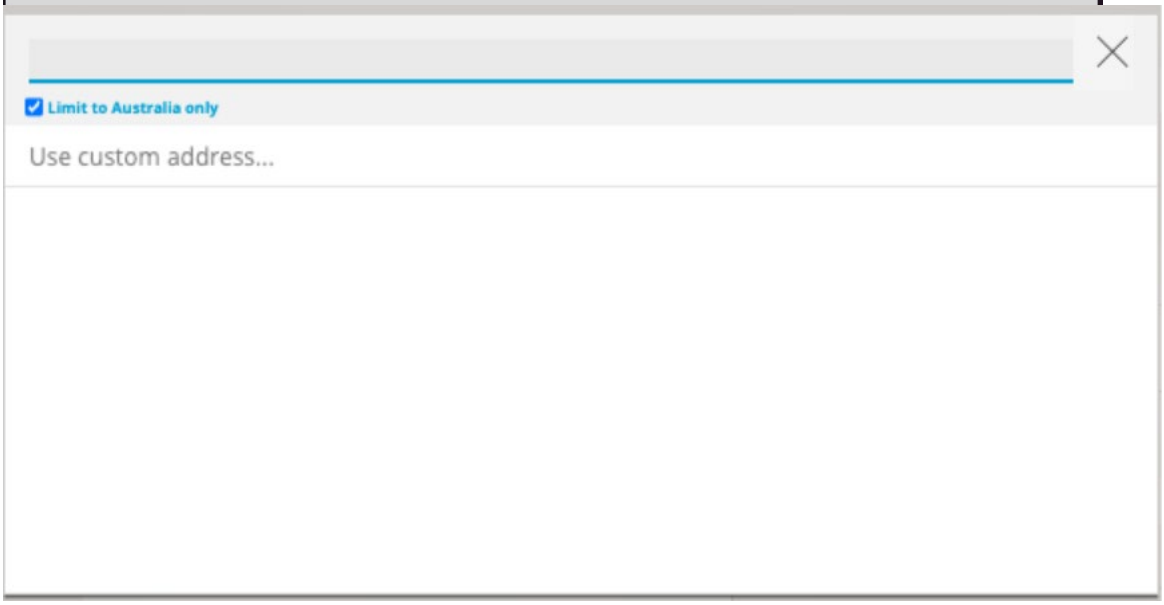
6.1.3 Site Access Deeds



6.2 Application Tab

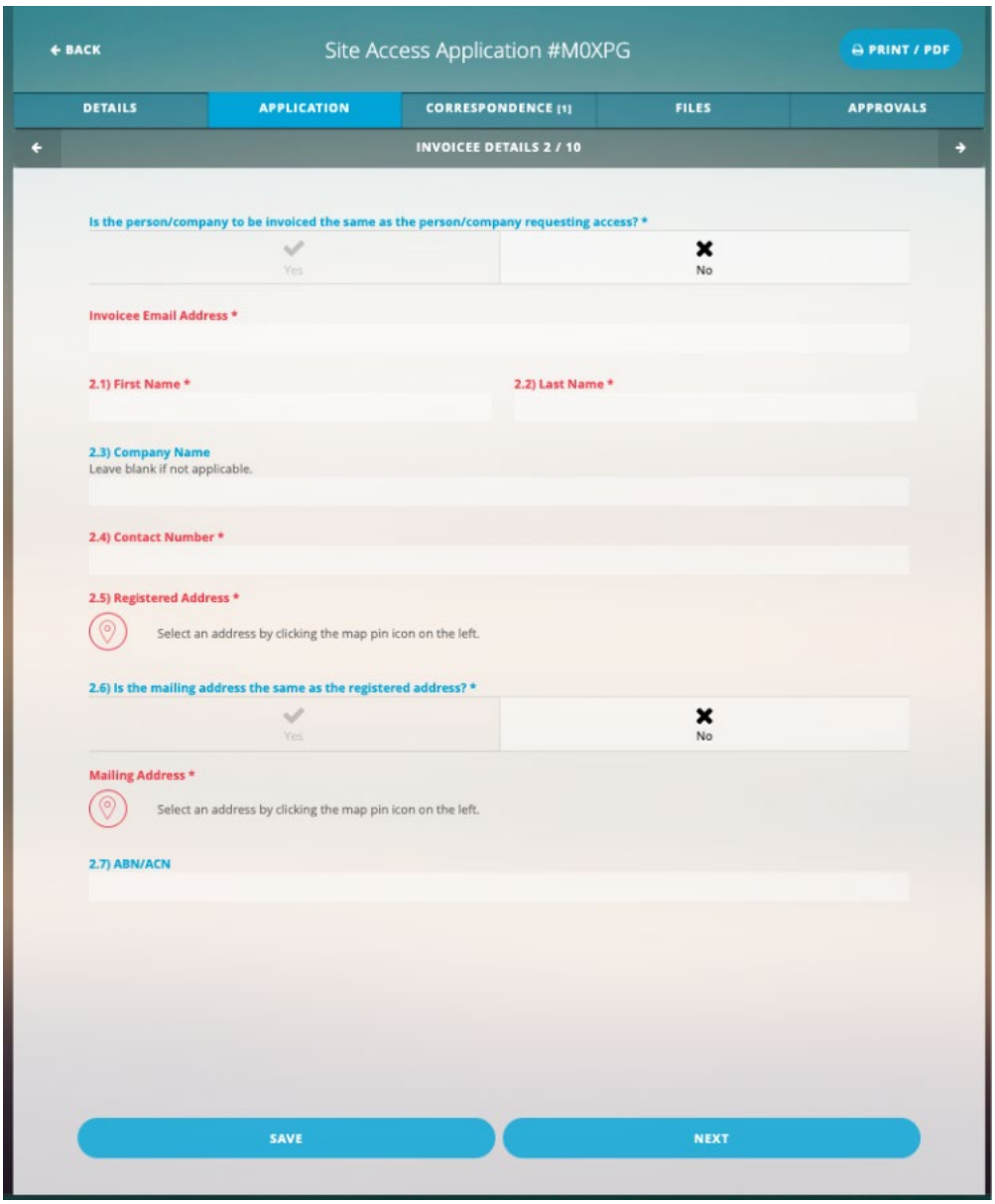
6.2.1 Application Details



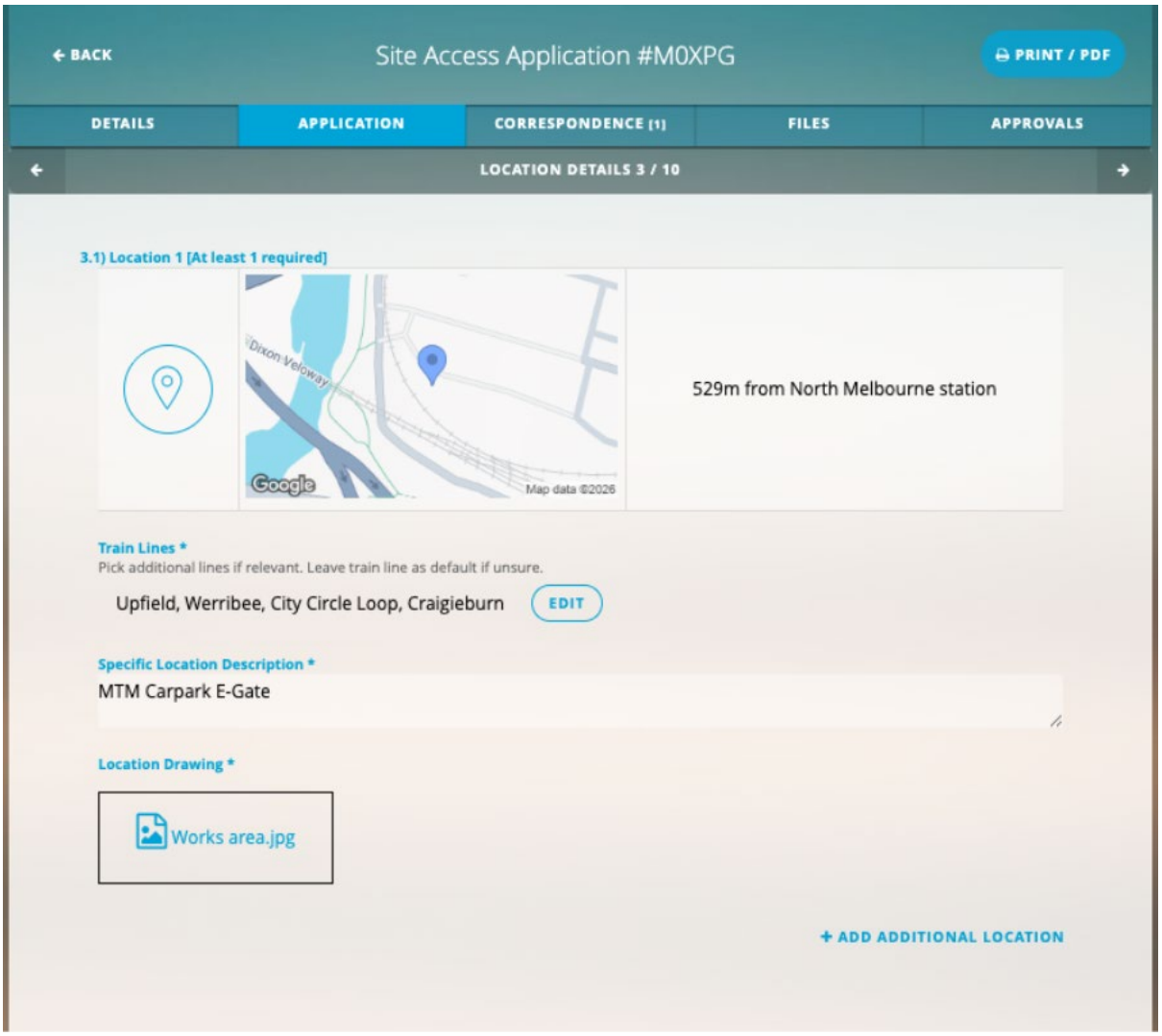


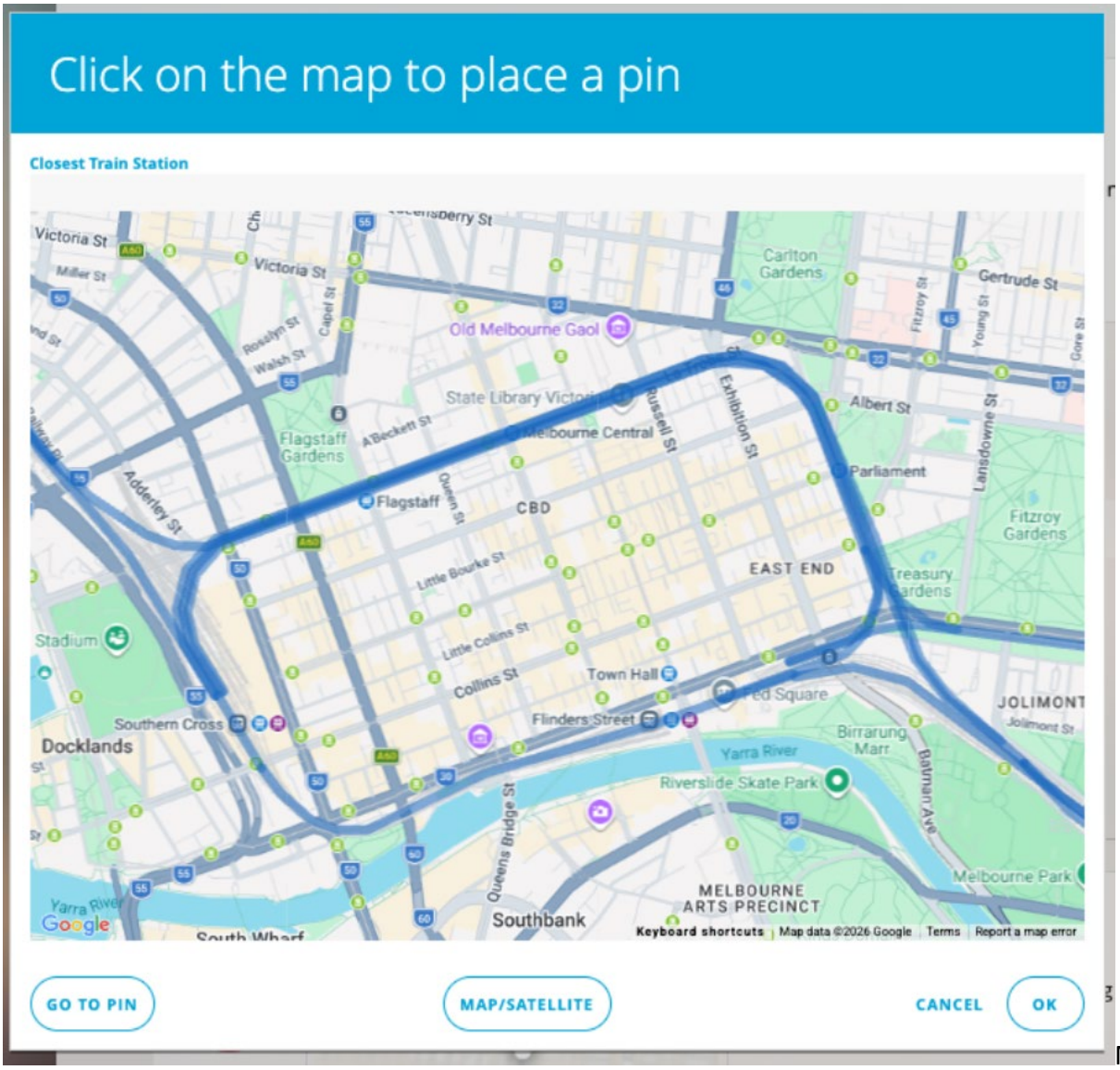
Address lookup.

6.2.2 Invoicee Details

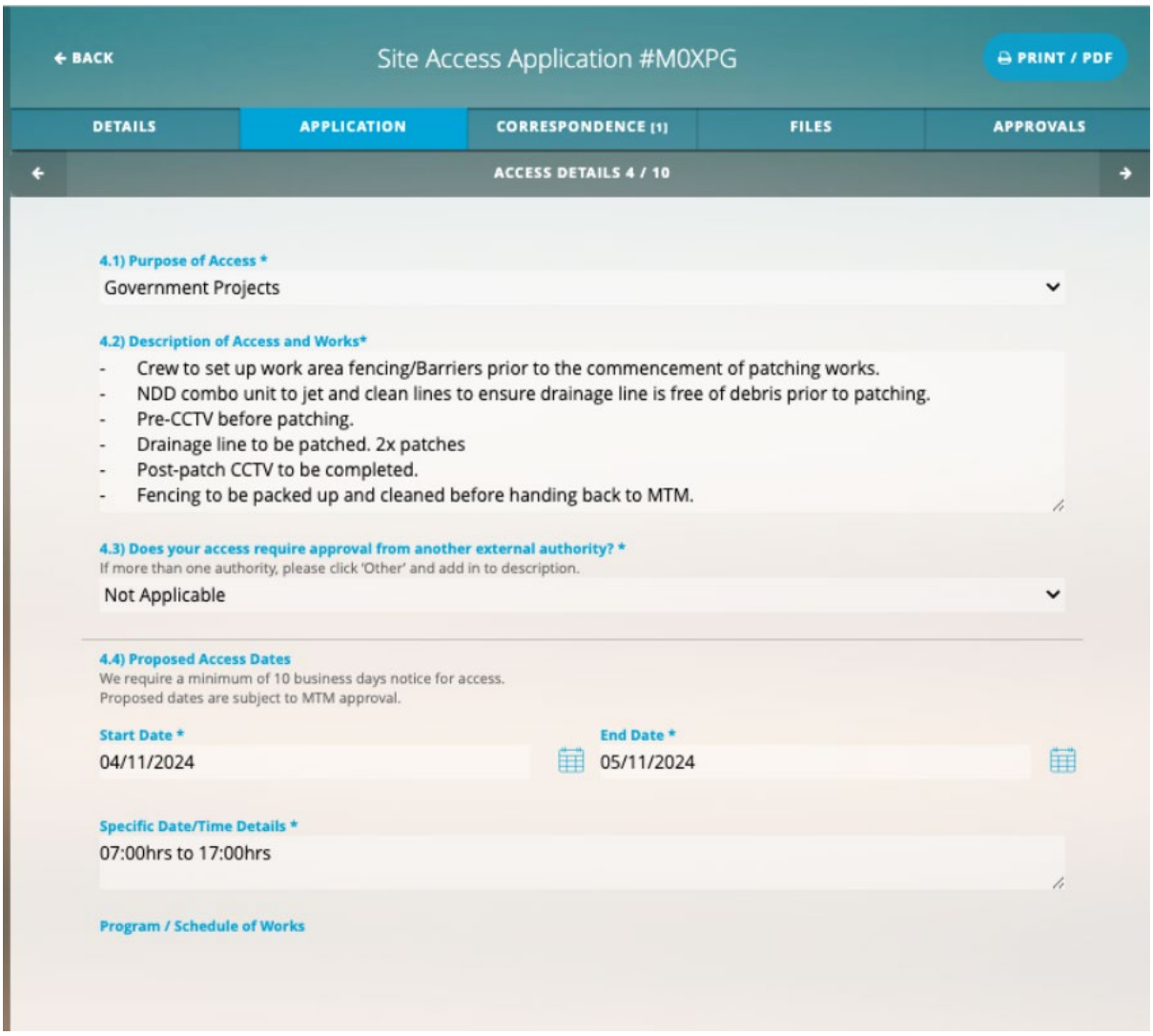


6.2.3 Location Details

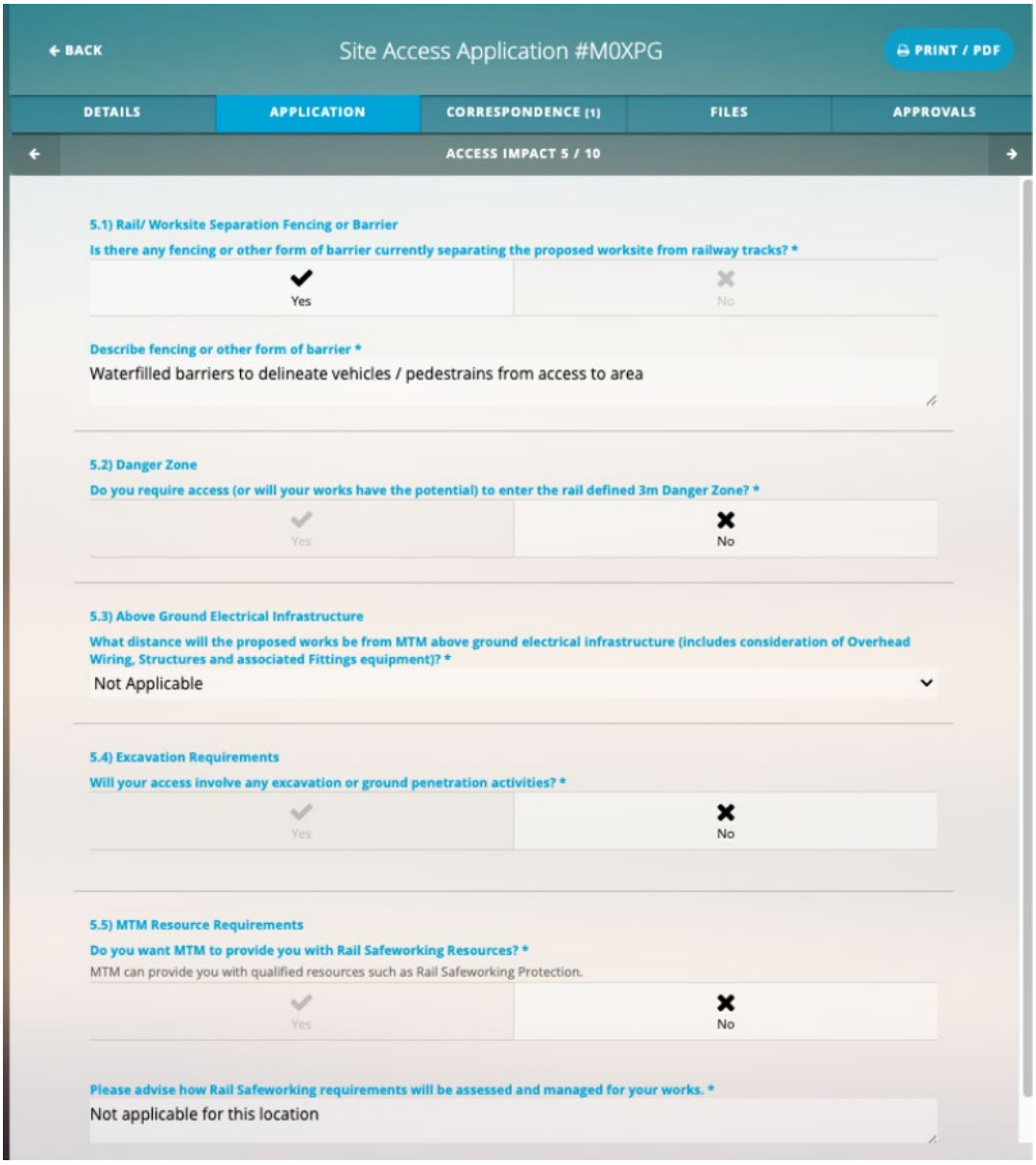



Click on the map to place a pin
Closest Train Station
Victoria St
Victoria St
Carlton Gardens
Gertrude St
Old Melbourne Gaol
State Library Victoria
Melbourne Central
Flagstaff Gardens
Flagstaff
CBD
Albert St
Parliament
Fitzroy Gardens
EAST END
Treasury Gardens
Stadium
Town Hall
Flinders Street
Southern Cross
Docklands
Yarra River
Riverslide Skate Park
Birrarung Marr
Melbourne Park
MELBOURNE ARTS PRECINCT
Southbank
Keyboard shortcuts
Map data ©2026 Google
Terms
Report a map error
GO TO PIN
MAP/SATELLITE
CANCEL
OK
Map Picker

6.2.4 Access Details



6.2.5 Access Impact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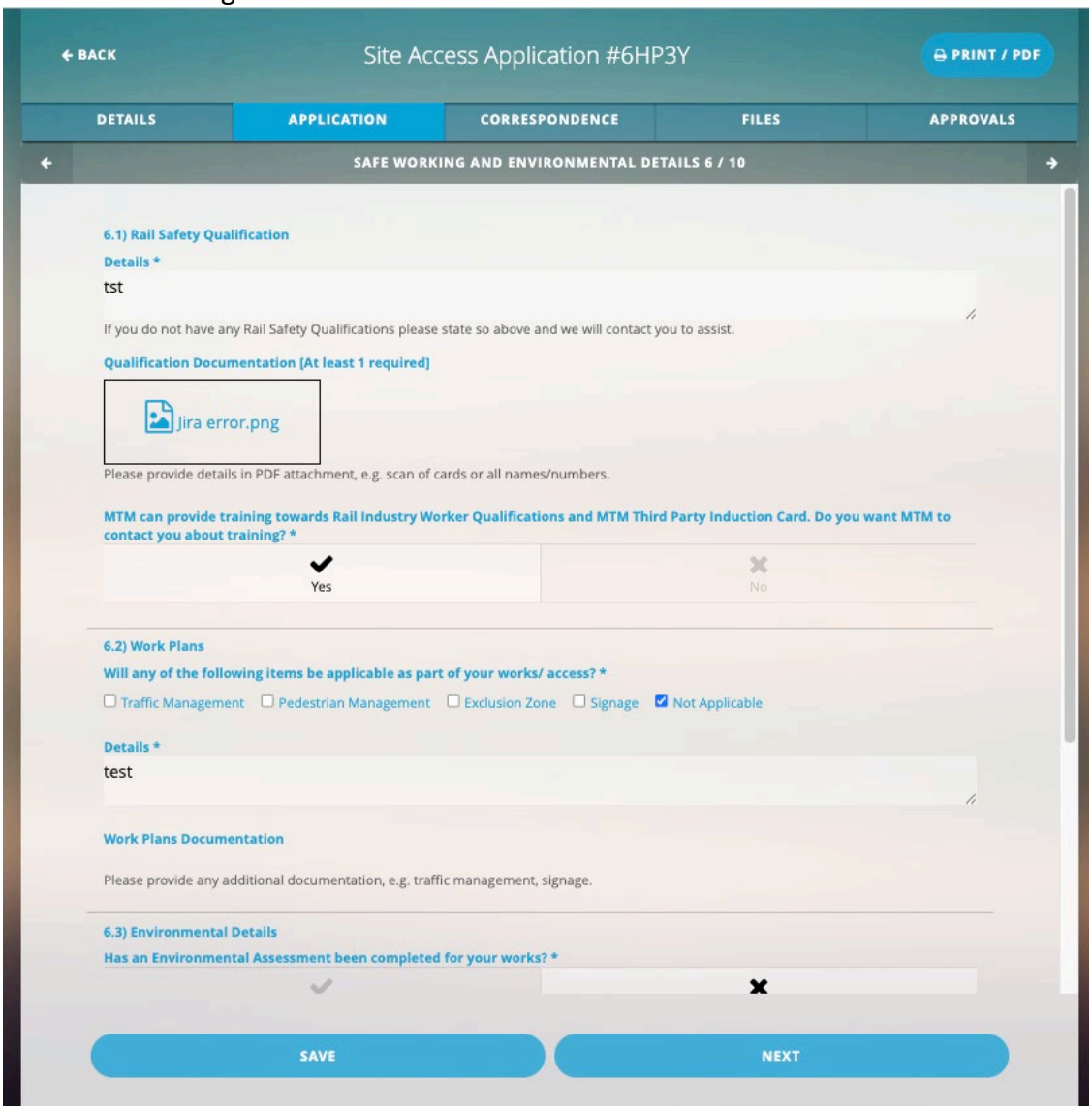
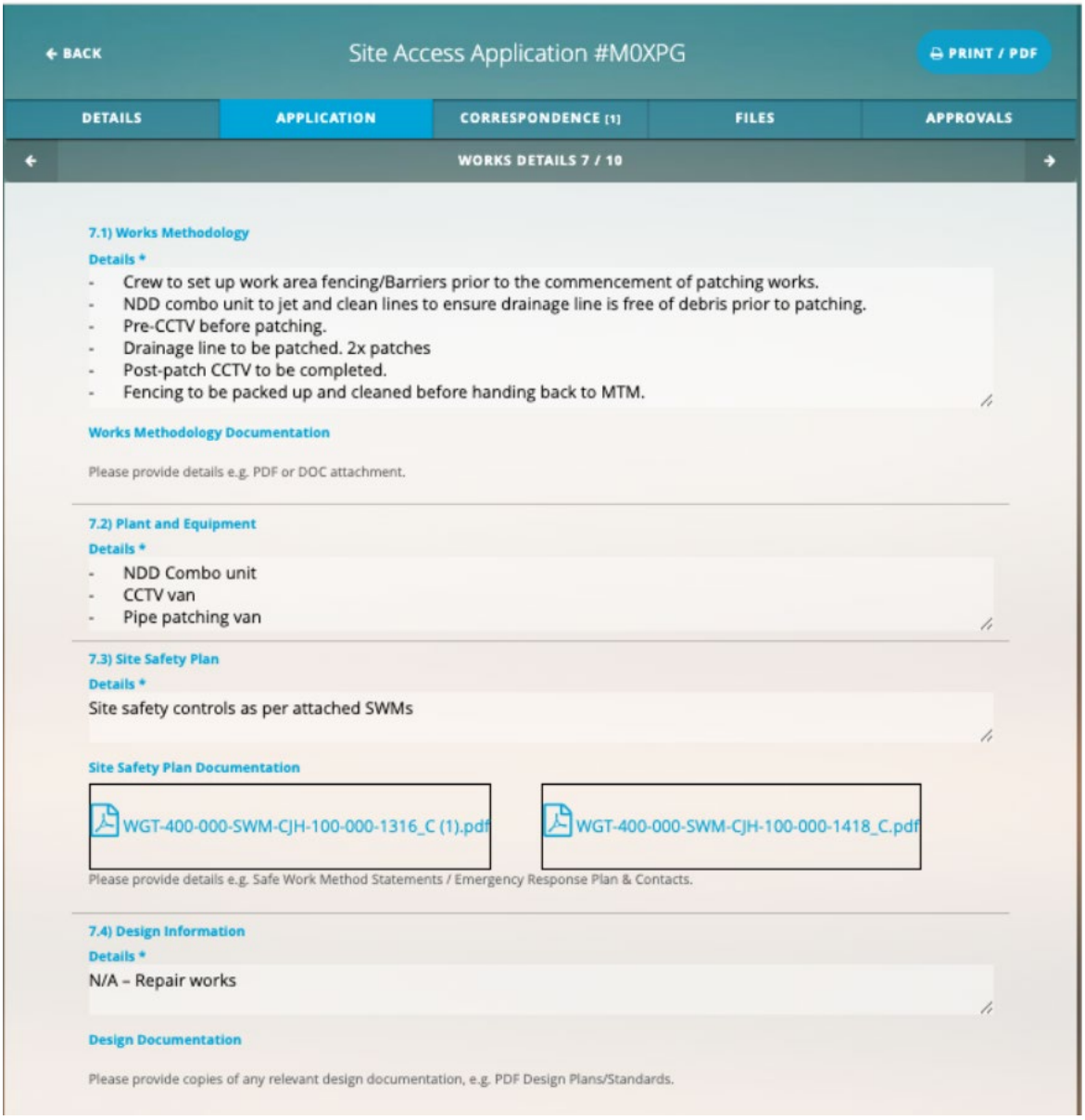
ACCESS IMPACT 5 / 10
5.1) Rail/ Worksite Separation Fencing or Barrier
Is there any fencing or other form of barrier currently separating the proposed worksite from railway tracks? *
Yes
No
Describe fencing or other form of barrier *
Waterfilled barriers to delineate vehicles / pedestrains from access to area
5.2) Danger Zone
Do you require access (or will your works have the potential) to enter the rail defined 3m Danger Zone? *
Yes
No
5.3) Above Ground Electrical Infrastructure
What distance will the proposed works be from MTM above ground electrical infrastructure (includes consideration of Overhead Wiring, Structures and associated Fittings equipment)? *
Not Applicable
5.4) Excavation Requirements
Will your access involve any excavation or ground penetration activities? *
Yes
No
5.5) MTM Resource Requirements
Do you want MTM to provide you with Rail Safeworking Resources? *
MTM can provide you with qualified resources such as Rail Safeworking Protection.
Yes
No
Please advise how Rail Safeworking requirements will be assessed and managed for your works. *
Not applicable for this location

6.2.6 Safe working and environmental Details



6.2.7 Work Details



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WORKS DETAILS 7 / 10

7.1) Works Methodology

Details *

- Crew to set up work area fencing/Barriers prior to the commencement of patching works.
- NDD combo unit to jet and clean lines to ensure drainage line is free of debris prior to patching.
- Pre-CCTV before patching.
- Drainage line to be patched. 2x patches
- Post-patch CCTV to be completed.
- Fencing to be packed up and cleaned before handing back to MTM.

Works Methodology Documentation

Please provide details e.g. PDF or DOC attachment.

7.2) Plant and Equipment

Details *

- NDD Combo unit
- CCTV van
- Pipe patching van

7.3) Site Safety Plan

Details *

Site safety controls as per attached SWMs

Site Safety Plan Documentation

WGT-400-000-SWM-CJH-100-000-1316_C (1).pdf

WGT-400-000-SWM-CJH-100-000-1418_C.pdf

Please provide details e.g. Safe Work Method Statements / Emergency Response Plan & Contacts.

7.4) Design Information

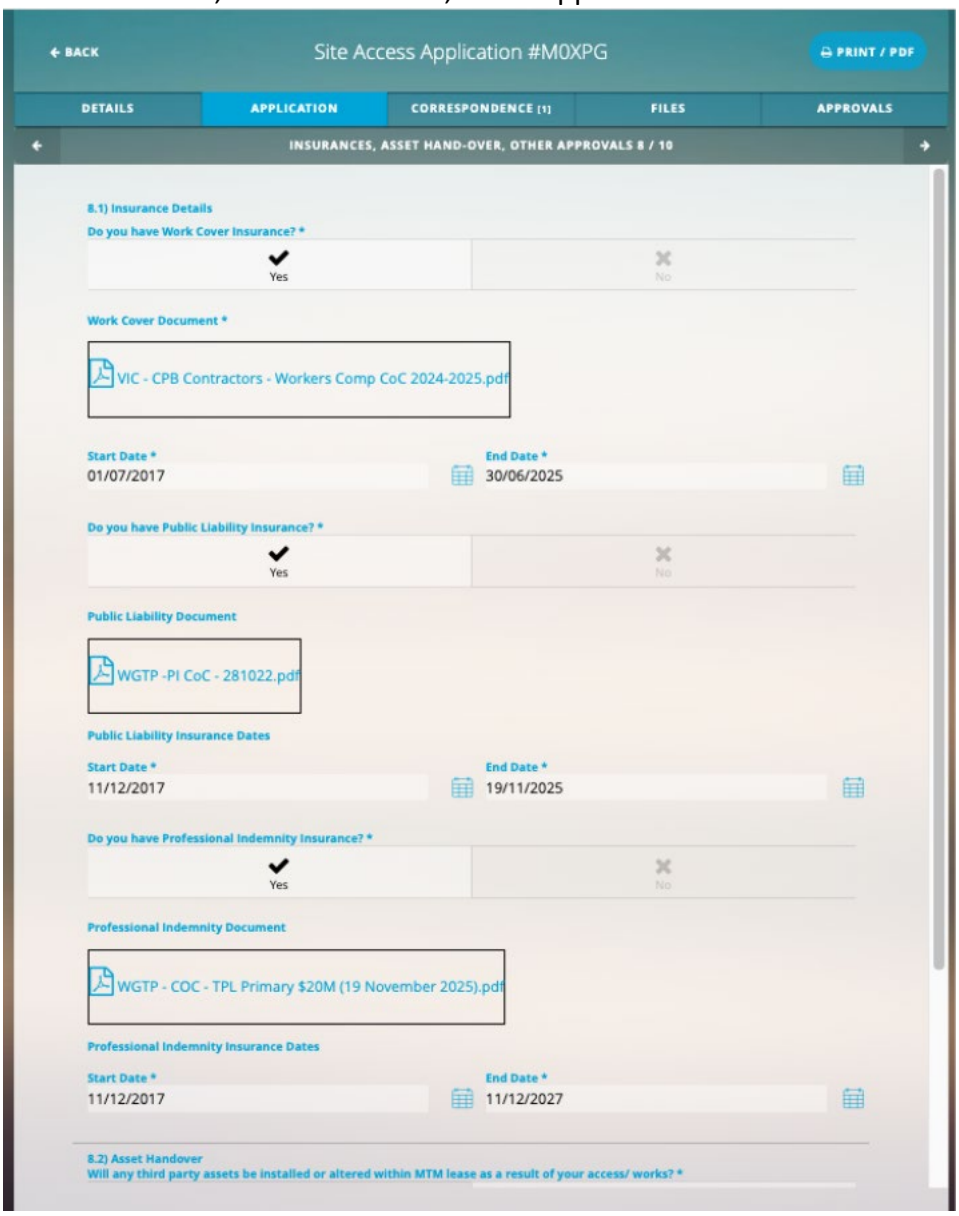
Details *

N/A – Repair works

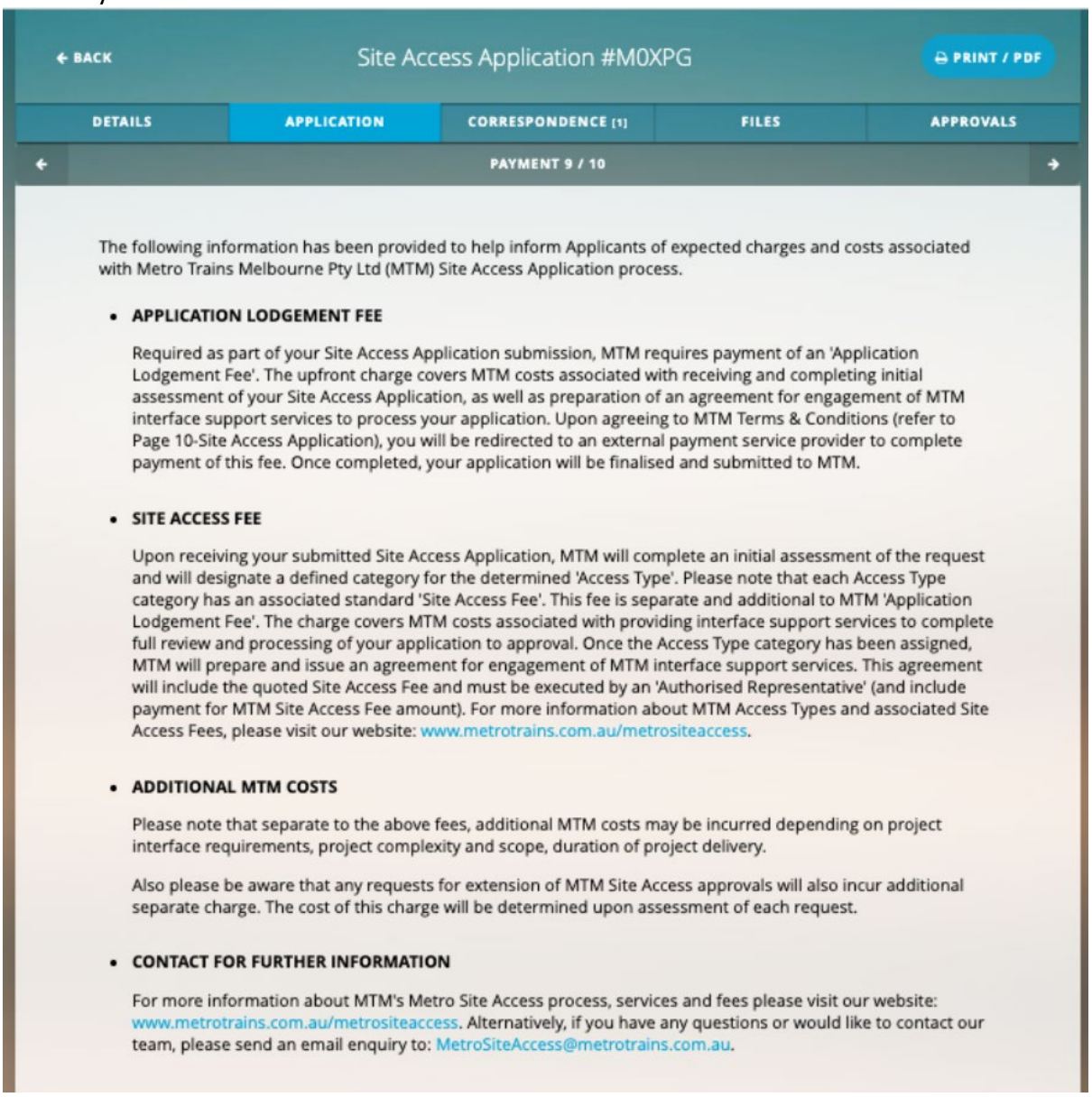
Design Documentation

Please provide copies of any relevant design documentation, e.g. PDF Design Plans/Standards.

6.2.8 Insurances, Asset Hand-over, other Approvals



6.2.9 Payment



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PAYMENT 9 / 10

The following information has been provided to help inform Applicants of expected charges and costs associated with Metro Trains Melbourne Pty Ltd (MTM) Site Access Application process.

- **APPLICATION LODGEMENT FEE**

  Required as part of your Site Access Application submission, MTM requires payment of an 'Application Lodgement Fee'. The upfront charge covers MTM costs associated with receiving and completing initial assessment of your Site Access Application, as well as preparation of an agreement for engagement of MTM interface support services to process your application. Upon agreeing to MTM Terms & Conditions (refer to Page 10-Site Access Application), you will be redirected to an external payment service provider to complete payment of this fee. Once completed, your application will be finalised and submitted to MTM.

- **SITE ACCESS FEE**

  Upon receiving your submitted Site Access Application, MTM will complete an initial assessment of the request and will designate a defined category for the determined 'Access Type'. Please note that each Access Type category has an associated standard 'Site Access Fee'. This fee is separate and additional to MTM 'Application Lodgement Fee'. The charge covers MTM costs associated with providing interface support services to complete full review and processing of your application to approval. Once the Access Type category has been assigned, MTM will prepare and issue an agreement for engagement of MTM interface support services. This agreement will include the quoted Site Access Fee and must be executed by an 'Authorised Representative' (and include payment for MTM Site Access Fee amount). For more information about MTM Access Types and associated Site Access Fees, please visit our website: www.metrotrains.com.au/metrositeaccess.

- **ADDITIONAL MTM COSTS**

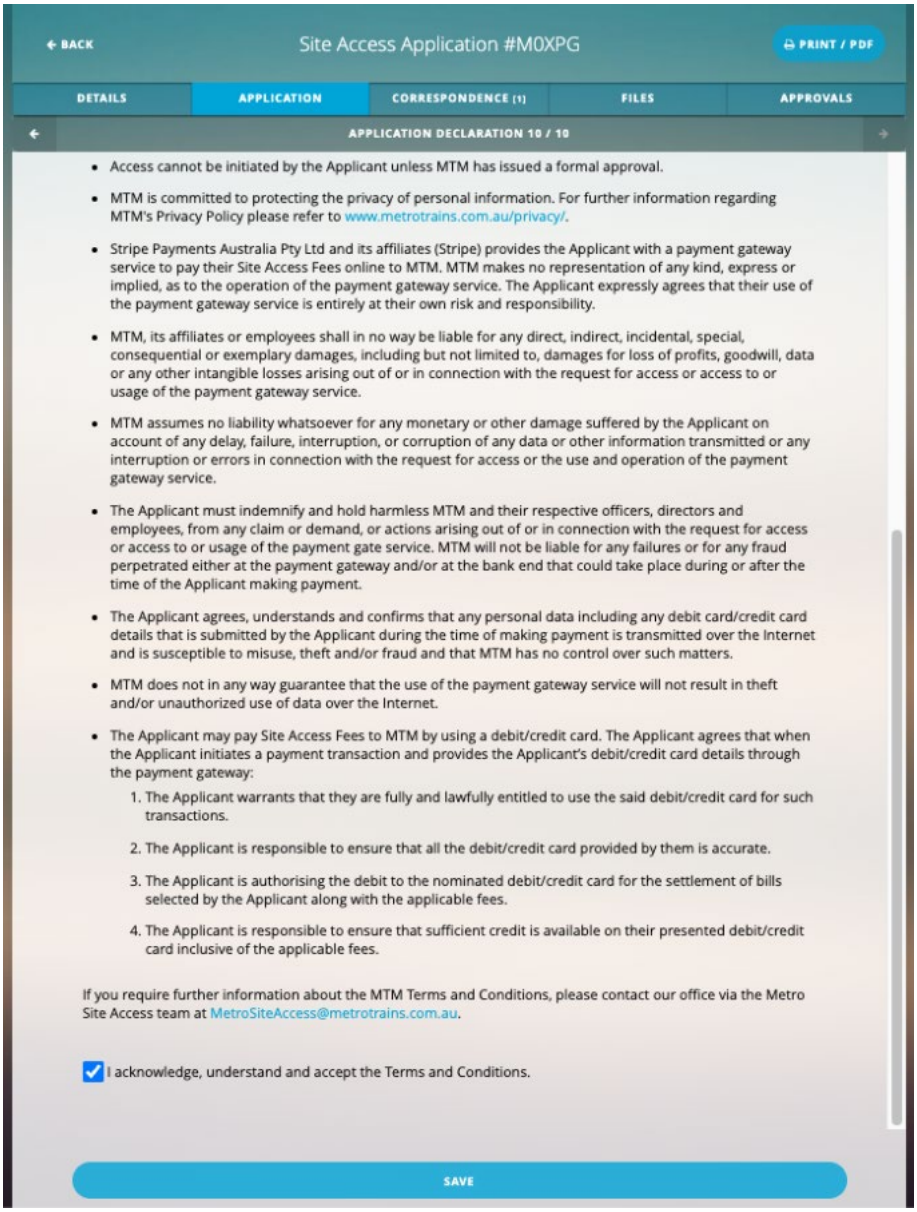
  Please note that separate to the above fees, additional MTM costs may be incurred depending on project interface requirements, project complexity and scope, duration of project delivery.

  Also please be aware that any requests for extension of MTM Site Access approvals will also incur additional separate charge. The cost of this charge will be determined upon assessment of each request.

- **CONTACT FOR FURTHER INFORMATION**

  For more information about MTM's Metro Site Access process, services and fees please visit our website: www.metrotrains.com.au/metrositeaccess. Alternatively, if you have any questions or would like to contact our team, please send an email enquiry to: MetroSiteAccess@metrotrains.com.au.

6.2.10 Acknowledgement



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APPLICATION DECLARATION 10 / 10

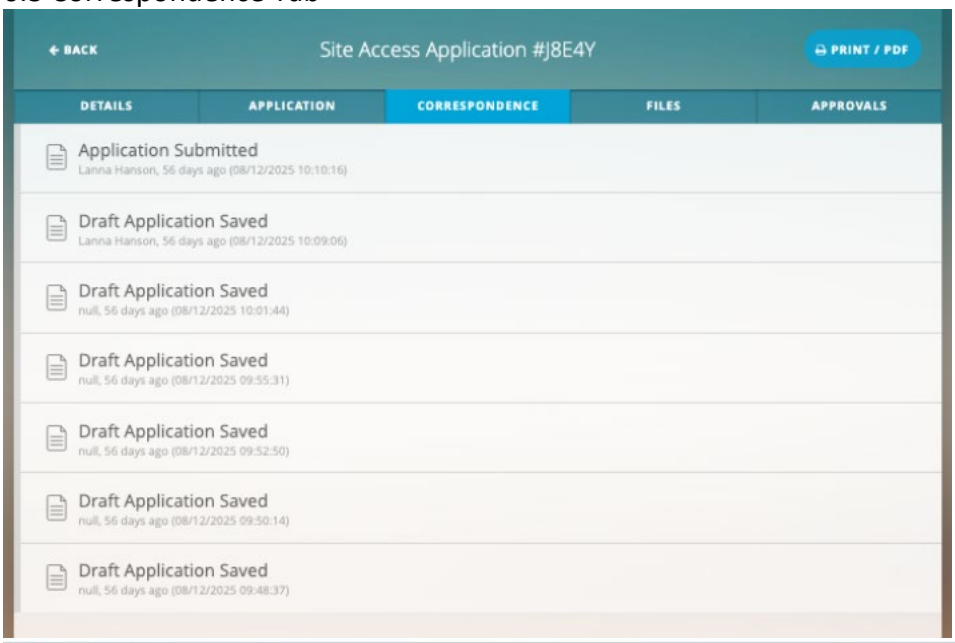
- Access cannot be initiated by the Applicant unless MTM has issued a formal approval.
- MTM is committed to protecting the privacy of personal information. For further information regarding MTM's Privacy Policy please refer to www.metrotrains.com.au/privacy/.
- Stripe Payments Australia Pty Ltd and its affiliates (Stripe) provides the Applicant with a payment gateway service to pay their Site Access Fees online to MTM. MTM makes no representation of any kind, express or implied, as to the operation of the payment gateway service. The Applicant expressly agrees that their use of the payment gateway service is entirely at their own risk and responsibility.
- MTM, its affiliates or employees shall in no way be liable for any direct, indirect, incidental, special, consequential or exemplary damages, including but not limited to, damages for loss of profits, goodwill, data or any other intangible losses arising out of or in connection with the request for access or access to or usage of the payment gateway service.
- MTM assumes no liability whatsoever for any monetary or other damage suffered by the Applicant on account of any delay, failure, interruption, or corruption of any data or other information transmitted or any interruption or errors in connection with the request for access or the use and operation of the payment gateway service.
- The Applicant must indemnify and hold harmless MTM and their respective officers, directors and employees, from any claim or demand, or actions arising out of or in connection with the request for access or access to or usage of the payment gate service. MTM will not be liable for any failures or for any fraud perpetrated either at the payment gateway and/or at the bank end that could take place during or after the time of the Applicant making payment.
- The Applicant agrees, understands and confirms that any personal data including any debit card/credit card details that is submitted by the Applicant during the time of making payment is transmitted over the Internet and is susceptible to misuse, theft and/or fraud and that MTM has no control over such matters.
- MTM does not in any way guarantee that the use of the payment gateway service will not result in theft and/or unauthorized use of data over the Internet.
- The Applicant may pay Site Access Fees to MTM by using a debit/credit card. The Applicant agrees that when the Applicant initiates a payment transaction and provides the Applicant's debit/credit card details through the payment gateway:
  1. The Applicant warrants that they are fully and lawfully entitled to use the said debit/credit card for such transactions.
  2. The Applicant is responsible to ensure that all the debit/credit card provided by them is accurate.
  3. The Applicant is authorising the debit to the nominated debit/credit card for the settlement of bills selected by the Applicant along with the applicable fees.
  4. The Applicant is responsible to ensure that sufficient credit is available on their presented debit/credit card inclusive of the applicable fees.

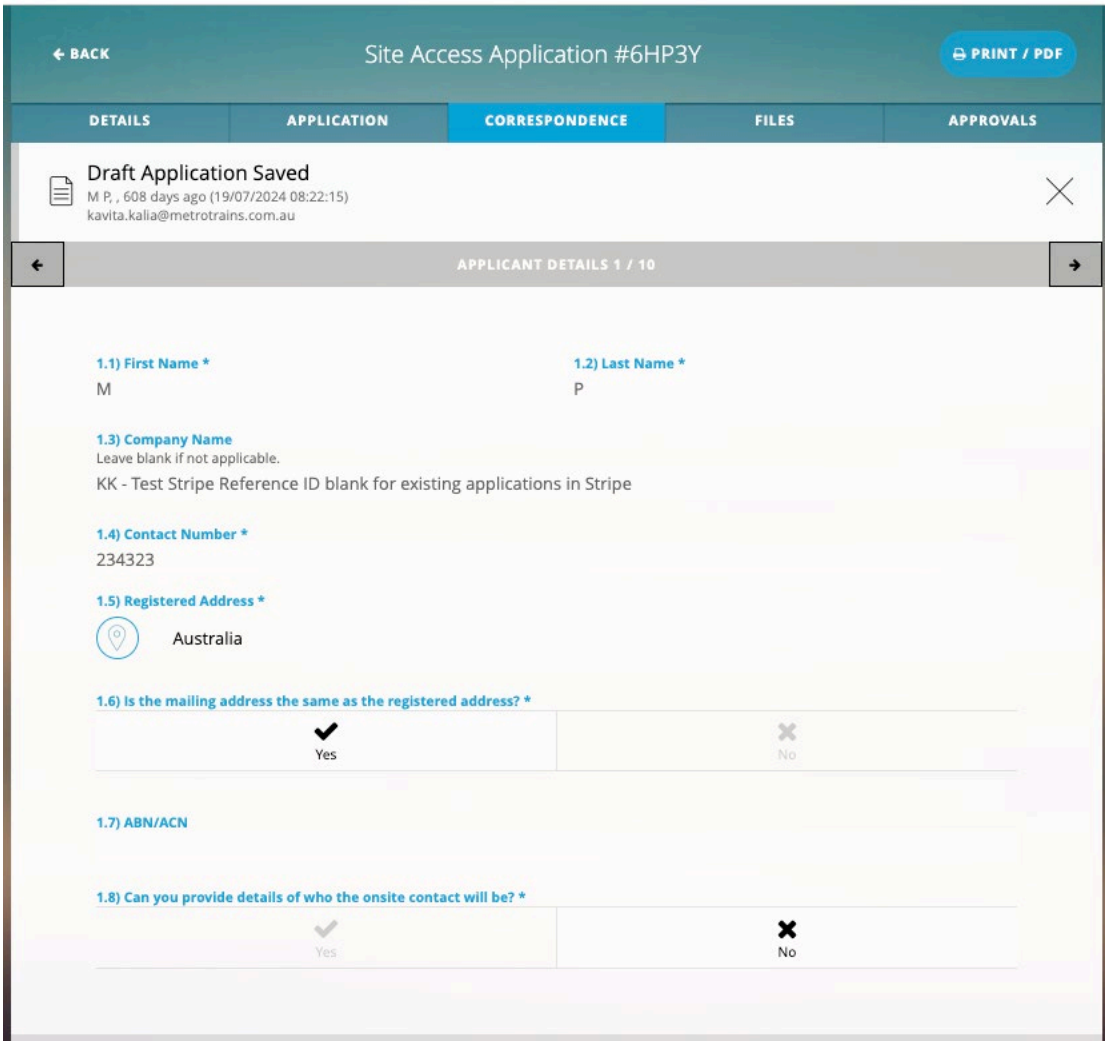
If you require further information about the MTM Terms and Conditions, please contact our office via the Metro Site Access team at MetroSiteAccess@metrotrains.com.au.

☑ I acknowledge, understand and accept the Terms and Conditions.

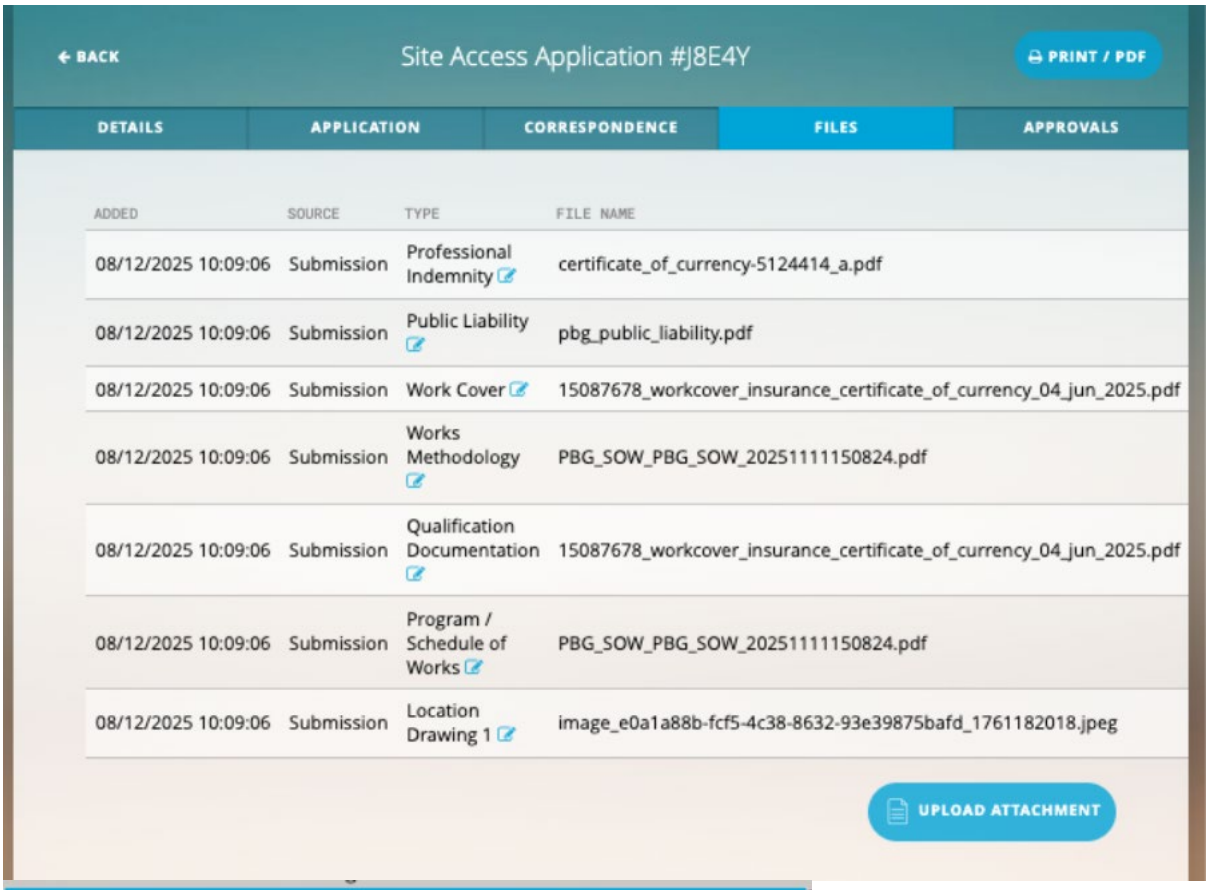
SAVE

6.3 Correspondence Tab





6.4 Files Tab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DETAILS
APPLICATION
CORRESPONDENCE
FILES
APPROVALS
ADDED
SOURCE
TYPE
FILE NAME
08/12/2025 10:09:06 Submission Professional Indemnity certificate_of_currency-5124414_a.pdf
08/12/2025 10:09:06 Submission Public Liability pbg_public_liability.pdf
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08/12/2025 10:09:06 Submission Program / Schedule of Works PBG_SOW_PBG_SOW_20251111150824.pdf
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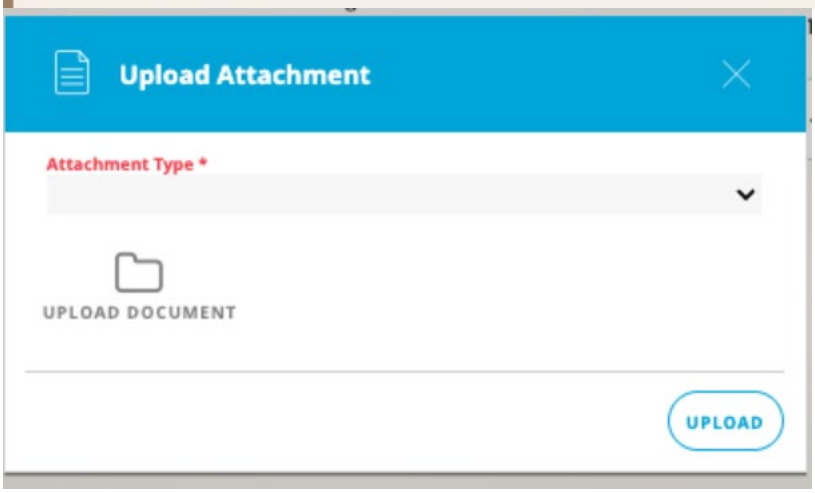
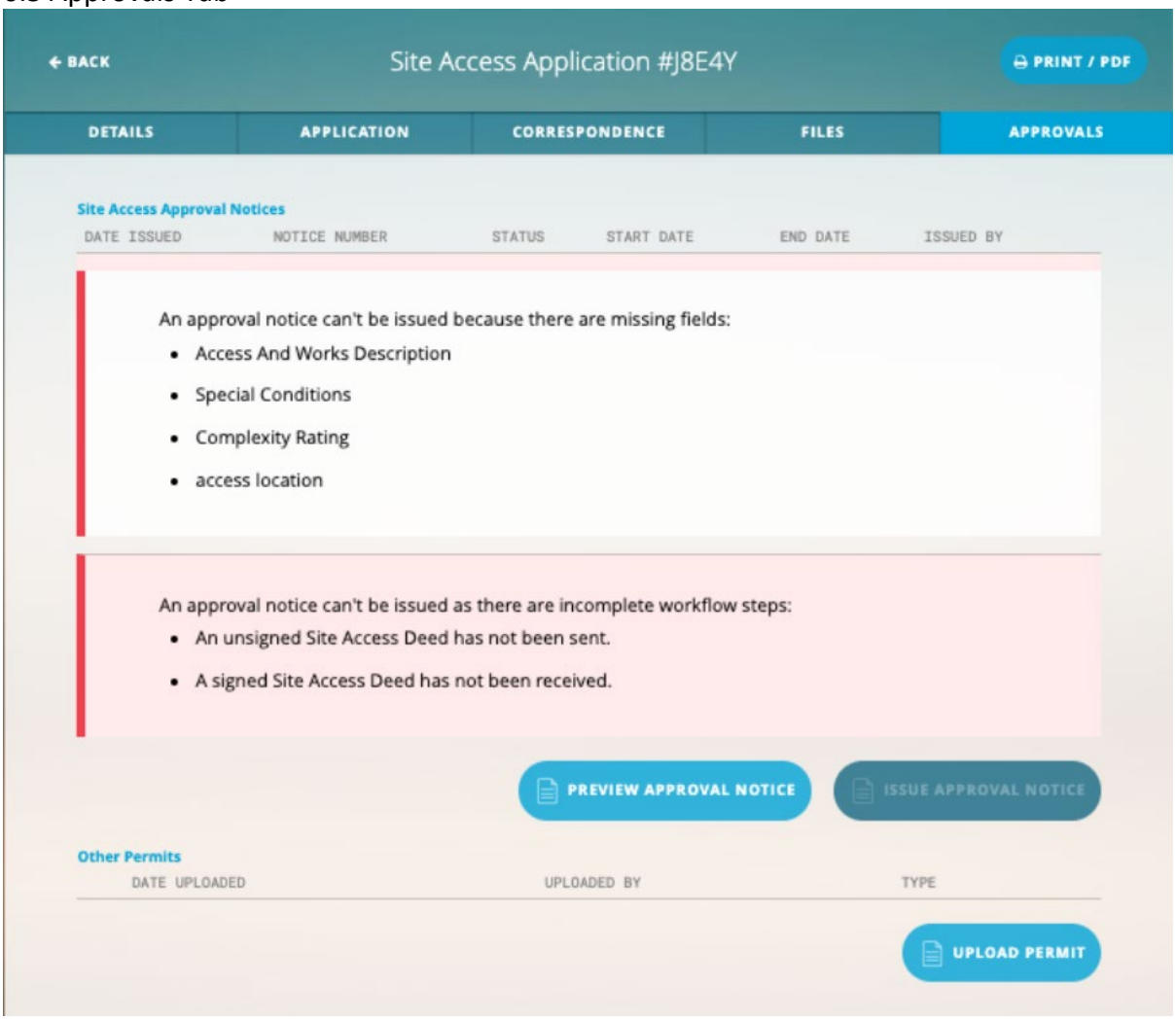
UPLOAD ATTACHMENT


Upload Attachment
Attachment Type *
UPLOAD DOCUMENT
UPLOAD

6.5 Approvals Tab



2.1.2. Scope Exclusions

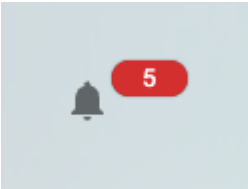
Items excluded from scope for change management are as follows:

No.	Out of Scope Item	Description
1.	Base project	Metro will provide a base project for the vendor to build out the relevant features
2.	Common Components	Metro will provide the core common UI components, i.e date pickers, style guide etc.
3.	Server backend changes	There are no changes to any server or backend components
4.	CI/CD	The existing Continuous Integration infrastructure and deployment process will be retained.
5.	IOS Apps	Migration from Cordova to Capacitor is no longer required

3. UI Common Components

The below is a list of common UI components that have been built as part of earlier migrations which should be used as part of this upgrade.

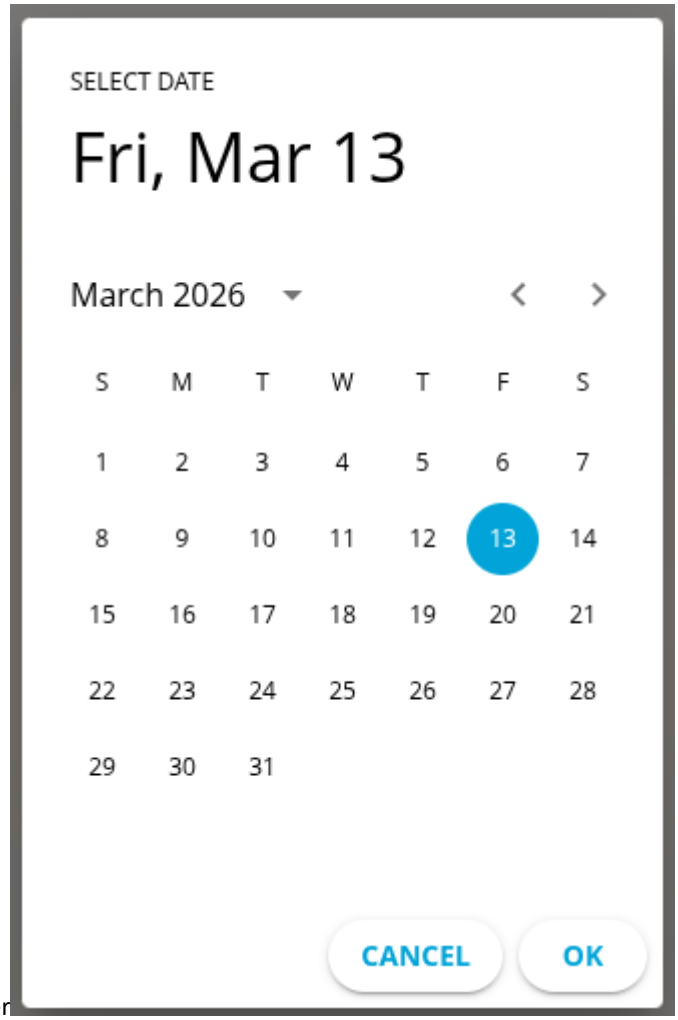
- Badge



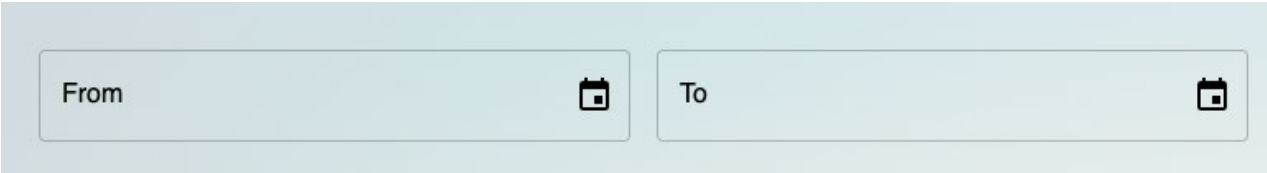
- Button

- Checkbox

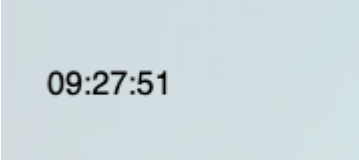
- Date Picker



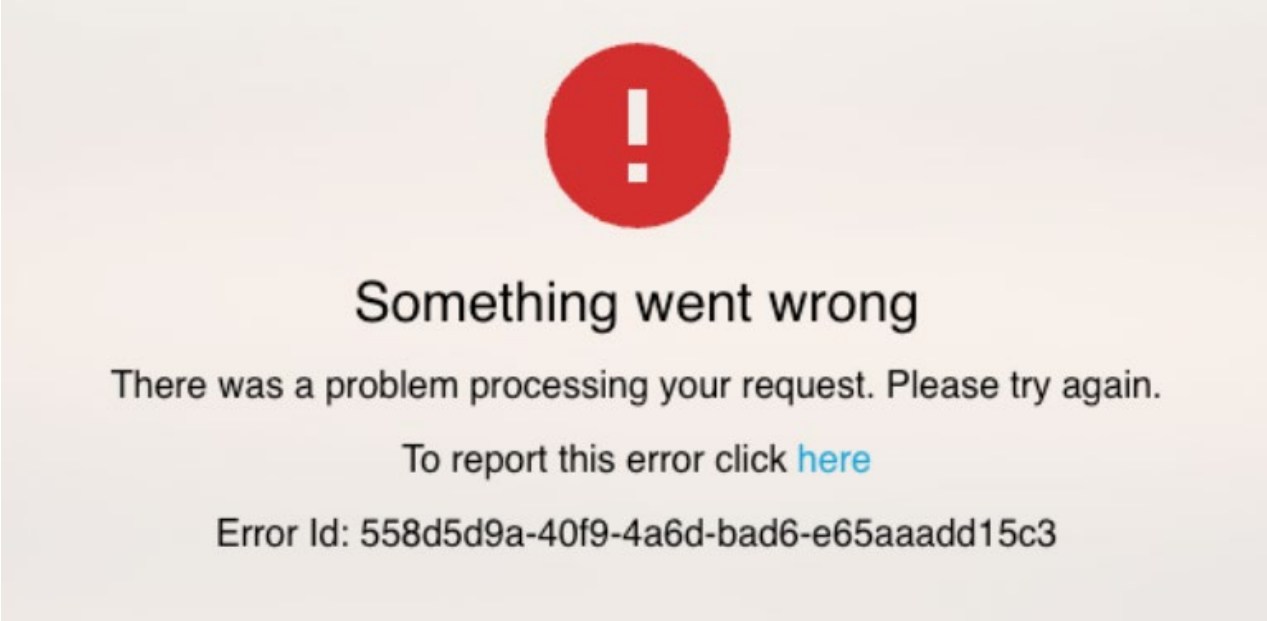
- Date Range Picker



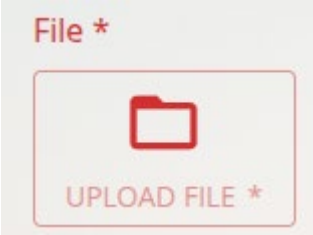
- Digital Clock



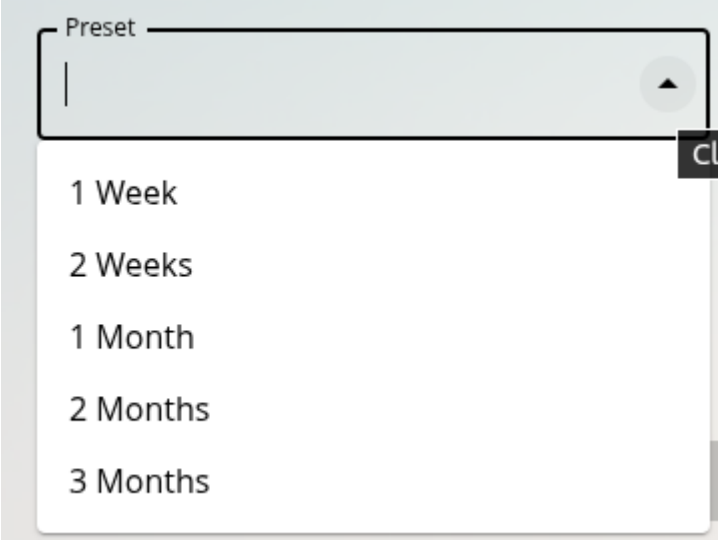
- Error Handler



- FileUpoloadButton

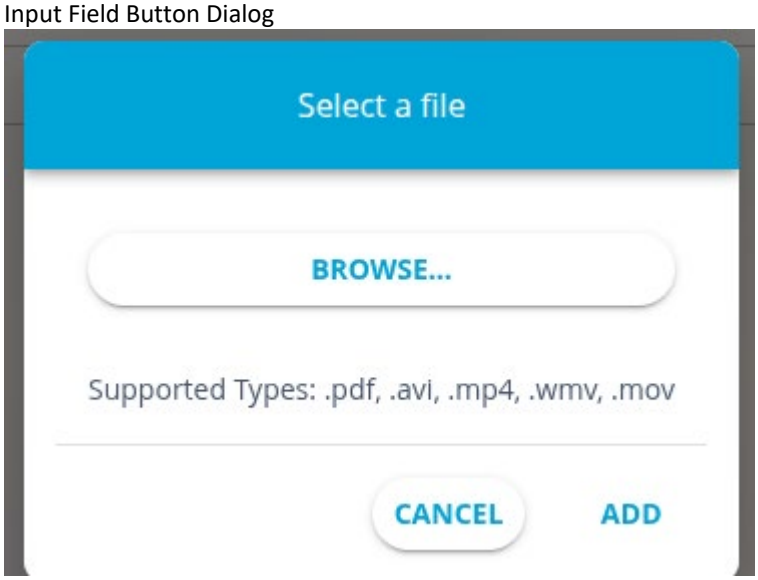


- Input Drop Down

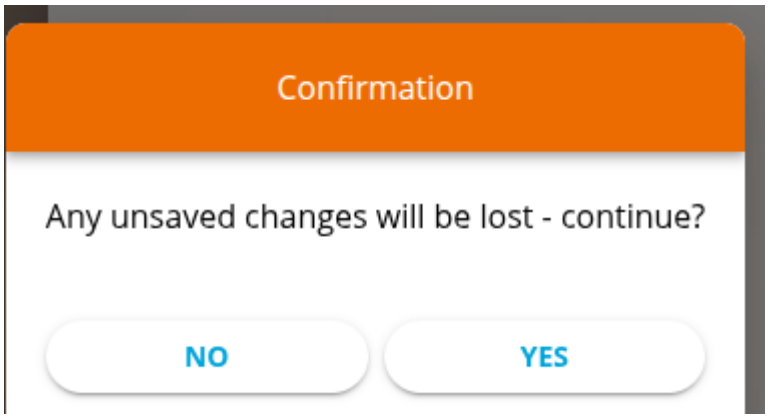


- Input Field

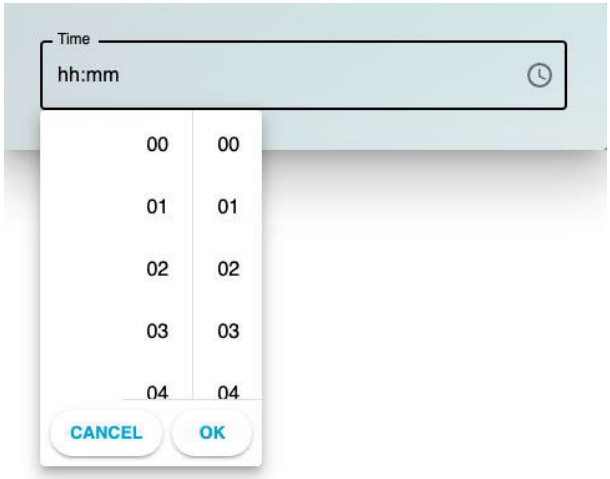
- Input Field Button Dialog



- Modal



- Time Picker



- Toast

4. Code Analysis

NLOC	Avg NLOC	Avg CCN	Avg Token	Function Count	file
519	9.8	1.8	60.2	57	./app/scripts/controllers/network-map.js
12	9	2	53	1	./app/scripts/controllers/home-chooser.js
31	5.3	1.7	27.3	6	./app/scripts/controllers/add-email-internal-dialogue.js
36	8.8	1.8	45	4	./app/scripts/controllers/create-user.js
265	7	1.4	41.8	43	./app/scripts/controllers/application-search.js
4	2	1	7	1	./app/scripts/controllers/menu.js
14	4	1	20.7	3	./app/scripts/controllers/find-by-application-number-dialogue.js
72	9.4	1	57.6	8	./app/scripts/controllers/home.js
116	8.4	1.9	48.7	15	./app/scripts/controllers/applications.js
15	7	1	38	2	./app/scripts/controllers/settings.js
18	4.2	1	22.5	4	./app/scripts/controllers/upload-email-attachment-dialogue.js
66	6.5	1.9	40.3	11	./app/scripts/controllers/issue-permit-dialogue.js
8	4	1	19	1	./app/scripts/controllers/user-admin.js
8	4	1	19	1	./app/scripts/controllers/help-and-support.js
34	5.8	1.5	36.2	6	./app/scripts/controllers/upload-file-attachment-dialogue.js
31	5.3	1.7	29.7	6	./app/scripts/controllers/update-email-address.js
153	30.8	3.4	145.8	5	./app/scripts/controllers/popup-occupation.js
66	5.7	1.2	32.2	13	./app/scripts/controllers/assessment.js
27	5.4	1.4	25.8	5	./app/scripts/controllers/add-email-external-dialogue.js
13	5	1	32	2	./app/scripts/controllers/list-users.js
34	5.8	1.3	33.7	6	./app/scripts/controllers/upload-other-permit-dialogue.js
92	5.6	1.4	32.8	16	./app/scripts/controllers/search-applications-dialogue.js
142	7	2	35.7	23	./app/scripts/controllers/add-finance-dialogue.js
68	22.7	2.3	162.7	3	./app/scripts/services/marker-gen.js
14	4.3	1	22	3	./app/scripts/services/create-user.js
49	6.8	1.9	29.4	8	./app/scripts/services/gmaploader.js
28	4	1	23.5	8	./app/scripts/services/site-access-application.js
16	5	1	31.7	3	./app/scripts/services/occupations.js
95	8.6	1.9	47.3	12	./app/scripts/services/route-history.js
71	4.9	1	30.2	17	./app/scripts/services/review.js
17	5.3	1.3	23.3	3	./app/scripts/services/settings.js
35	4.9	1	34.2	8	./app/scripts/services/finance.js
9	4	1	10.5	2	./app/scripts/services/applicant.js
32	5	1	32.3	7	./app/scripts/services/correspondence.js
35	4.9	1	35.2	8	./app/scripts/services/attachments.js
35	7.4	1.8	46.6	5	./app/scripts/services/common-locations.js

15	4	1	24.2	4	./app/scripts/services/network.js
10	4.5	1	15	2	./app/scripts/directives/menu-button.js
42	11	2.5	58.2	4	./app/scripts/directives/money.js
21	6.3	1.3	27	3	./app/scripts/directives/assessment-application.js
118	6.5	1.5	39.4	19	./app/scripts/directives/assessment-attachments.js
17	5.7	1	24.7	3	./app/scripts/directives/map.js
424	6.7	1.6	41.3	73	./app/scripts/directives/assessment-details.js
71	12.3	3.5	81.2	6	./app/scripts/directives/location-description.js
270	7.9	1.9	45.8	38	./app/scripts/directives/assessment-correspondence.js
132	6.3	1.2	35.2	24	./app/scripts/directives/assessment-approvals.js
95	74	1	279	1	./app/scripts/app.js
444	8.5	2.6	65.1	52	./app/scripts/markermanager.js
171	5.7	1.8	33.4	32	./app/scripts/app-firstrun.js
15	5.3	1	26.7	3	./test/spec/controllers/create-user.js
15	5.3	1	26.7	3	./test/spec/controllers/application-search.js
15	5.3	1	26.7	3	./test/spec/controllers/home.js
15	5.3	1	26.7	3	./test/spec/controllers/applications.js
15	5.3	1	26.7	3	./test/spec/controllers/user-admin.js
15	5.3	1	26.7	3	./test/spec/controllers/help-and-support.js
15	5.3	1	26.7	3	./test/spec/controllers/search-applications-dialogue.js.js
15	5.3	1	26.7	3	./test/spec/controllers/update-email-address.js
15	5.3	1	26.7	3	./test/spec/controllers/popup-occupation.js
15	5.3	1	26.7	3	./test/spec/controllers/assessment.js
15	5.3	1	26.7	3	./test/spec/controllers/list-users.js
15	5.3	1	26.7	3	./test/spec/controllers/homechooser.js
15	5.3	1	26.7	3	./test/spec/controllers/add-finance-dialogue.js
11	4	1	19	3	./test/spec/services/create-user.js
11	4	1	19	3	./test/spec/services/occupations.js
11	4	1	19	3	./test/spec/services/review.js
11	4	1	19	3	./test/spec/services/settings.js
11	4	1	19	3	./test/spec/services/finance.js
11	4	1	19	3	./test/spec/services/applicant.js
11	4	1	19	3	./test/spec/services/correspondence.js
11	4	1	19	3	./test/spec/services/attachments.js
14	5	1	29.3	3	./test/spec/directives/money.js
14	5	1	29.3	3	./test/spec/directives/assessment-application.js
14	5	1	29.3	3	./test/spec/directives/assessment-attachments.js

5. Code samples

The below is an example of a directive build in AngularJS.

```
'use strict';

/**
 * @ngdoc directive
 * @name uiApp.directive:assessmentApplication
 * @description
 * # assessmentApplication
 */
angular.module('uiApp')
.directive('assessmentApplication', ['$log', 'Review',
    function ($log, Review) {
        return {
            templateUrl: 'views/assessment-application.html',
            restrict: 'E',
            scope: {
                model: '=',
                showing: '='
            },
            link: function postLink(scope, elementUnused, attrsUnused) {
                scope.$watch('showing', function (newValue, oldValue) {
//                    $log.info('New value of showing is ' + newValue);
                    if (true === newValue) {
                        Review.attentionGiven(scope.model.applicationNumber);
                    }
                });
            }
        };
    }
]);
```

6. Libraries and dependencies

Library	Description
vite	A fast, modern build tool with instant server start and optimized production builds.
axios	A promise-based HTTP client for making API requests with interceptors and error handling.
typescript	A typed superset of JavaScript that helps catch errors early and improves code quality.

Library	Description
zustand	A simple and unopinionated state management library for React with minimal boilerplate.
lodash	A utility library providing helpful functions for working with arrays, objects, and strings.
date-fns	A comprehensive toolset for manipulating and formatting dates in JavaScript.
materialui	A popular React UI framework that implements Google's Material Design principles.
tailwindcss	A utility-first CSS framework for rapidly building custom designs.
storybook	A development environment for building and testing UI components in isolation.
vitest	A fast unit testing framework designed for Vite projects with a Jest-like API.
jest	A robust JavaScript testing framework with features like mocking and snapshot testing.